1 A block of weight 50 N is in equilibrium, suspended from fixed points A and B which are 2 m apart on a horizontal ceiling.

Fig. 7.1 illustrates one way of doing this. A light, inextensible string of length 2.8 m is passed round a small smooth light pulley attached to a point C on the block. The parts of the string from C to A and from C to B should be treated as straight lines making angles θ and ϕ with the vertical.

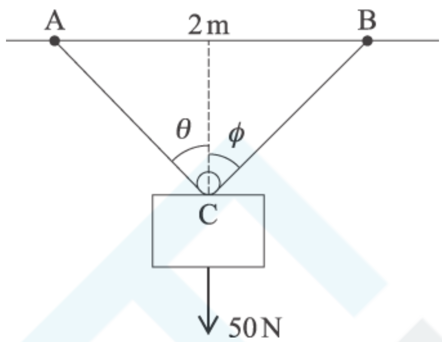


Fig. 7.1

(i)

A State which piece of the information that you have been given tells you that the tension in the string is the same on each side of the pulley.

[1]

B Hence show that θ = ϕ.

[2]

(ii) Show that $\cos\theta = \frac{\sqrt{24}}{7}$.

[2]

(iii) Find the tension in the string.

[3]

Fig. 7.2 illustrates another way of suspending the block from the same two points, A and B, with the string now cut into two parts, AC and BC. The length of AC is 1.2 m and BC is 1.6 m. The angles the strings make with the horizontal are α and β. The tension in the string AC is T_1 N and the tension in the string BC is T_2 N.

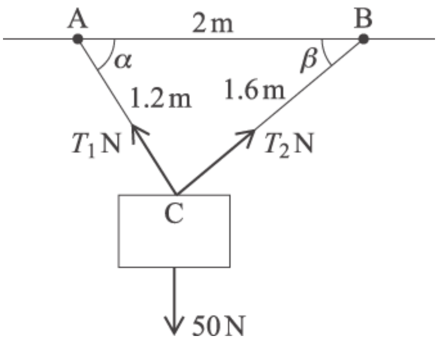


Fig. 7.2

(iv) Show that ∠ACB = 90°.

Write down the values of cos α and cos β.

[2]

(v) Find T_1 and T_2.

[5]

In a different arrangement, the string is cut so that the lengths of the two parts are 0.5 m and 2.3 m.

(vi) Describe how the block hangs in equilibrium in this case and state the tensions in the two strings.

[3]

2 Fig. 1 shows a block of mass 3 kg on a plane which is inclined at an angle of 30° to the horizontal.

A force P N is applied to the block parallel to the plane in the upwards direction.

The plane is rough so that a frictional force of 10 N opposes the motion.

The block is moving at constant speed up the plane.

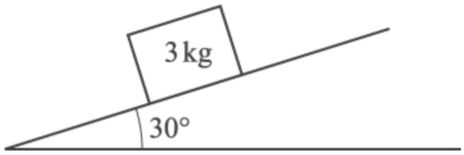


Fig. 1

(i) Mark and label all the forces acting on the block.

[3]

(ii) Calculate the magnitude of the normal reaction of the plane on the block.

[1]

(iii) Calculate the magnitude of the force P.

[2]

3 Fig. 8.1 shows a sledge of mass 40 kg. It is being pulled across a horizontal surface of deep snow by a light horizontal rope. There is a constant resistance to its motion.

The tension in the rope is 120 N.

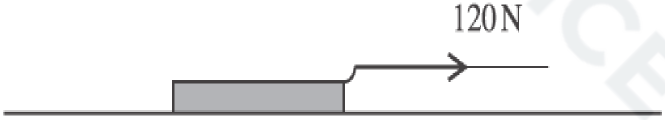


Fig. 8.1

The sledge is initially at rest. After 10 seconds its speed is 5 ms^{-1}.

(i) Show that the resistance to motion is 100 N. **[4]**

When the speed of the sledge is 5 ms^{-1}, the rope breaks.

The resistance to motion remains 100 N.

(ii) Find the speed of the sledge

A 1.6 seconds after the rope breaks, **[3]**

B 6 seconds after the rope breaks. **[1]**

The sledge is then pushed to the bottom of a ski slope. This is a plane at an angle of 15° to the horizontal.

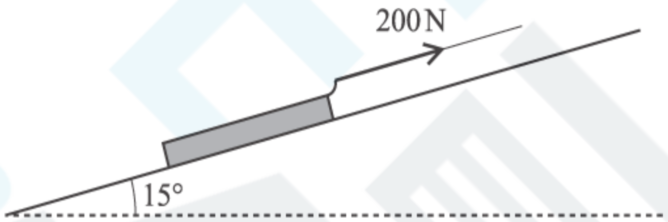


Fig. 8.2

The sledge is attached by a light rope to a winch at the top of the slope. The rope is parallel to the slope and has a constant tension of 200 N. Fig. 8.2 shows the situation when the sledge is part of the way up the slope.

The ski slope is smooth.

(iii) Show that when the sledge has moved from being at rest at the bottom of the slope to the point when its speed is 8 ms^{-1}, it has travelled a distance of 13.0 m (to 3 significant figures). **[4]**

When the speed of the sledge is 8 ms^{-1}, this rope also breaks.

(iv) Find the time between the rope breaking and the sledge reaching the bottom of the slope. **[6]**

4 Abi and Bob are standing on the ground and are trying to raise a small object of weight 250 N to the top of a building. They are using long light ropes. Fig. 7.1 shows the initial situation.

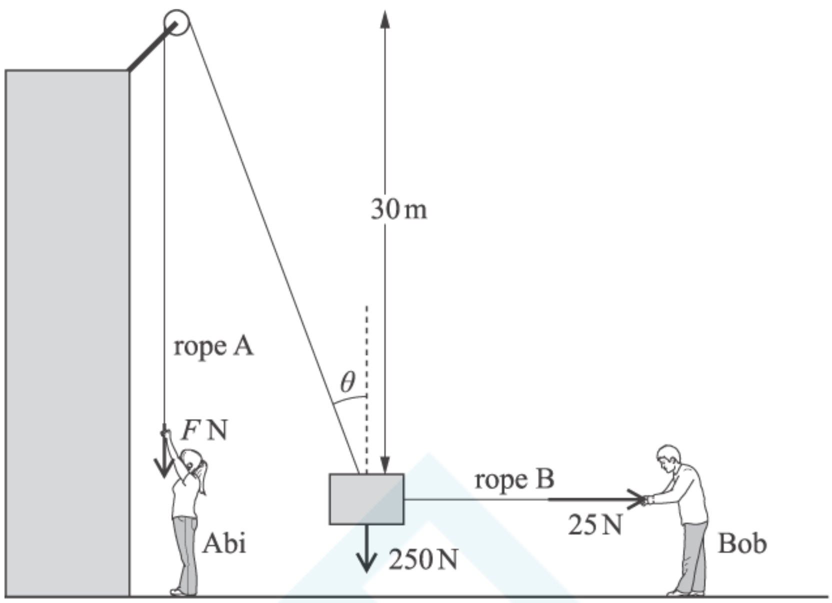


Fig. 7.1

Abi pulls vertically downwards on the rope A with a force F N. This rope passes over a small smooth pulley and is then connected to the object. Bob pulls on another rope, B, in order to keep the object away from the side of the building.

In this situation, the object is stationary and in equilibrium. The tension in rope B, which is horizontal, is 25 N. The pulley is 30 m above the object. The part of rope A between the pulley and the object makes an angle θ with the vertical.

(i) Represent the forces acting on the object as a fully labelled triangle of forces. **[3]**

(ii) Find F and θ.

Show that the distance between the object and the vertical section of rope A is 3 m. **[5]**

Abi then pulls harder and the object moves upwards. Bob adjusts the tension in rope B so that the object moves along a vertical line.

Fig. 7.2 shows the situation when the object is part of the way up. The tension in rope A is S N and the tension in rope B is T N. The ropes make angles α and β with the vertical as shown in the diagram. Abi and Bob are taking a rest and holding the object stationary and in equilibrium.

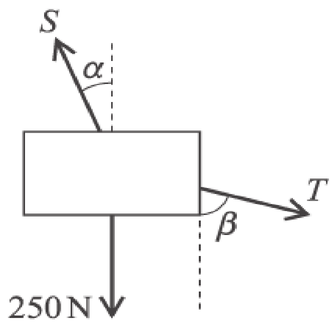


Fig. 7.2

(iii) Give the equations, involving S, T, α and β, for equilibrium in the vertical and horizontal directions. [3]

(iv) Find the values of S and T when α = 8.5° and β = 35°. [4]

(v) Abi's mass is 40 kg.

Explain why it is not possible for her to raise the object to a position in which α = 60°. [3]

5 Fig. 1 shows a pile of four uniform blocks in equilibrium on a horizontal table. Their masses, as shown, are 4 kg, 5 kg, 7 kg and 10 kg.

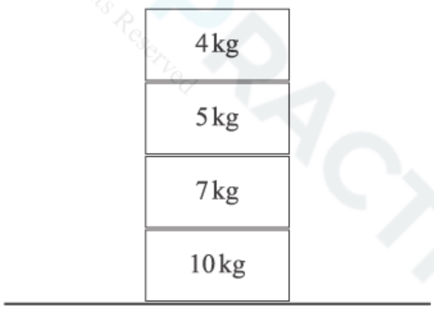


Fig.1

Mark on the diagram the magnitude and direction of each of the forces acting on the 7 kg block. [3]

6 Fig. 3 shows a smooth ball resting in a rack. The angle in the middle of the rack is 90°. The rack has one edge at angle α to the horizontal.

The weight of the ball is W N. The reaction forces of the rack on the ball at the points of contact are R N and S N.

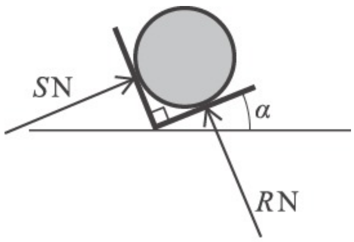


Fig. 3

(i) Draw a fully labelled triangle of forces to show the forces acting on the ball. Your diagram must indicate which angle is α.

[3]

(ii) Find the values of R and S in terms of W and α.

[2]

(iii) On the same axes draw sketches of R against α and S against α for $0° \leqslant \alpha \leqslant 90°$.

For what values of α is $R < S$?

[3]

7 Fig. 2 shows a 6 kg block on a smooth horizontal table. It is connected to blocks of mass 2 kg and 9 kg by two light strings which pass over smooth pulleys at the edges of the table. The parts of the strings attached to the 6 kg block are horizontal.

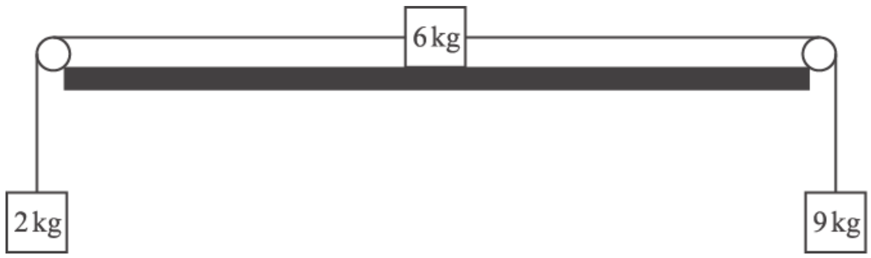


Fig. 2

(i) Draw three separate diagrams showing all the forces acting on each of the blocks.

[3]

(ii) Calculate the acceleration of the system and the tension in each string.

[5]

8 Fig. 1 shows four forces acting at a point. The forces are in equilibrium.

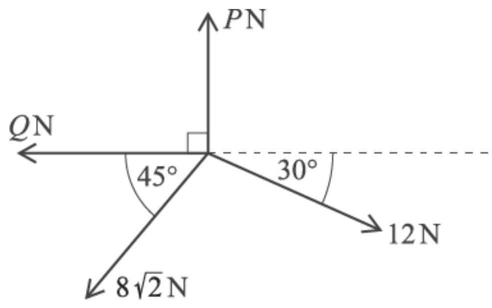


Fig. 1

Show that $P = 14$.

Find Q, giving your answer correct to 3 significant figures.

[5]

9 Fig. 7 illustrates a situation on a building site. An unexploded bomb is being lifted by light ropes that pass over smooth pulleys. The ropes are attached to winches V and W.

- The weight of the bomb is 7500 N.
- The winches are on horizontal ground and are at the same level.
- The sloping parts of the ropes from V and W are at angles α and β to the horizontal.
- The point P is level with the horizontal sections of the ropes and is 16 m and 9 m from the two pulleys, as shown.
- The winches are controlled so that the bomb moves in a vertical line through P. The tension in the rope attached to winch W is kept constant at 8000 N. The tension, T N, in the rope attached to winch V is varied.
- The distance between the top of the bomb, B, and the point P is d metres.

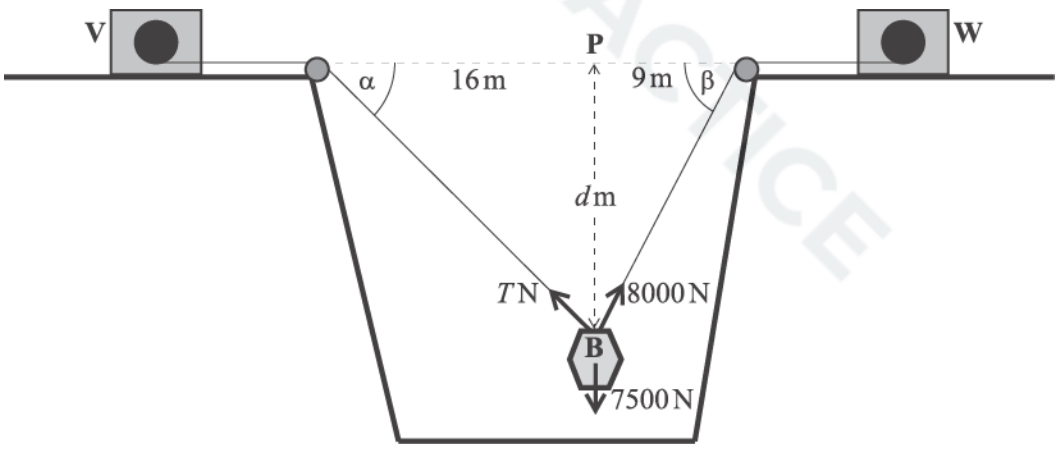


Fig. 7

At a particular stage in the lift, d = 12 and T = 6000.

(i) Find the values of $\cos\alpha$ and $\cos\beta$ at this stage.

[1]

(ii) Verify that, at this stage, the horizontal component of the bomb's acceleration is zero. Find the vertical component of its acceleration.

[7]

At a later stage, the bomb is higher up and so the values of d, T, α and β have all changed.

(iii) Show that $T = \dfrac{8000\cos\beta}{\cos\alpha}$.

Hence show that $T = \dfrac{4500\sqrt{d^2+256}}{\sqrt{d^2+81}}$.

[4]

(iv) Find the acceleration of the bomb when d = 6.75.

[4]

(v) Explain briefly why it is not possible for the bomb to be in equilibrium with B at P.

What could you say about the acceleration of the bomb if B were at P and the tensions in the two ropes were equal?

[2]

10 Fig. 1 shows a block of mass M kg being pushed over level ground by means of a light rod. The force, T N, this exerts on the block is along the line of the rod.

The ground is rough.

The rod makes an angle a with the horizontal.

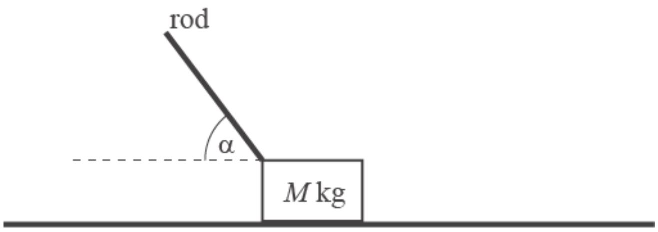


Fig. 1

(i) Draw a diagram showing all the forces acting on the block.

[3]

(ii) You are given that $M = 5$, $\alpha = 60°$, $T = 40$ and the acceleration of the block is $1.5\ \text{ms}^{-2}$.

Find the frictional force.

[3]

11 Fig. 3 shows a particle of weight 8 N on a rough horizontal table. The particle is being pulled by a horizontal force of 10 N. It remains at rest in equilibrium.

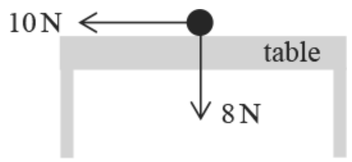


Fig. 3

(a) What information given in the question tells you that the forces shown in Fig. 3 cannot be the only forces acting on the particle? **[1]**

(b) The only other forces acting on the particle are due to the particle being on the table. State the types of these forces and their magnitudes. **[2]**

12 A small package hangs from a balloon by means of a light inelastic string. The string is always vertical. The mass of the package is 15 kg.

Catherine initially models the situation by assuming that there is no air resistance to the motion of the package. Use Catherine's model to calculate the tension in the string if

(a) the package is held at rest by the tension in the string, **[1]**

(b) the package is instantaneously at rest and accelerating **upwards** at 2 m s^{-2}, **[2]**

(c) the package is moving **downwards** at 3 m s^{-1} and accelerating **upwards** at 2 m s^{-2}. **[1]**

Catherine now carries out an experiment to find the magnitude of the air resistance on the package when it is moving. At a time when the package is accelerating **downwards** at 1.5 m s^{-2}, she finds that the tension in the string is 140 N.

(d) Calculate the magnitude of the air resistance at that time. Give, with a reason, the direction of motion of the package. **[5]**

13 A woman is pulling a loaded sledge along horizontal ground. The only resistance to motion of the sledge is due to friction between it and the ground.

Fig. 5

At first she pulls with a force of 100N inclined at 32° to the horizontal, as shown in Fig.5, but the sledge does not move.

(a) Determine the frictional force between the ground and the sledge.
Give your answer correct to 3 significant figures. **[2]**

(b) Next she pulls with a force of 100N inclined at a smaller angle to the horizontal. The sledge still does not move.

[2]

Compare the frictional force in this new situation with that in part **(a)**, justifying your answer.

14 In an experiment a small box is hit across a floor. After it has been hit, the box slides without rotation. The box passes a point A. The distance the box travels after passing A before coming to rest is S metres and the time this takes is T seconds. The only resistance to the box's motion is friction due to the floor.

The mass of the box is m kg and the frictional force is a constant F N.

(a) (i) Find the equation of motion for the box while it is sliding.

(ii) Show that $S = kT^2$ where $k = \frac{F}{2m}$. **[4]**

(b) Given that $k = 1.4$, find the value of the coefficient of friction between the box and the floor. **[4]**

15 A block of mass $5m$ kg is in equilibrium on a rough horizontal table. It is connected by horizontal light inextensible strings over smooth pulleys to particles of mass m kg and $2m$ kg which hang freely, as shown in Fig. 3.

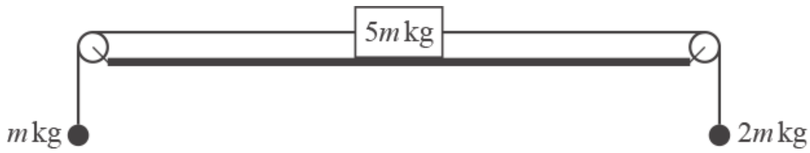


Fig. 3

Find the frictional force acting on the block, clearly indicating its direction. **[3]**

16 Dorothy and Harold push a wheelbarrow of mass 20 kg. They both push in the direction of motion. The wheelbarrow moves 10m from rest on a straight horizontal path. Dorothy applies a force of 90 N. Harold applies a force of 90 N for the first 3 m, and then reduces the force to 80 N for the remaining 7 m. The resistance to motion is 170 N throughout.

Find the time taken for the wheelbarrow to move the 10 m. **[9]**

17 Two chess pieces are placed on a uniform straight ruler. The ruler balances horizontally on a pivot.

- The ruler AB is of length 30 cm.
- The pivot P is at the centre of the ruler.
- The first chess piece, of mass 20 grams, is at A.
- The second chess piece, of mass 50 grams, is x cm from B.

This is shown in Fig. 5.

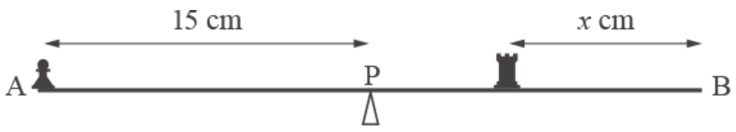


Fig. 5

Calculate the value of x. **[4]**

18 A globe hangs on the end of a light wire that makes an angle of θ with the vertical. It is held in place by a light horizontal string, as shown in Fig. 7.

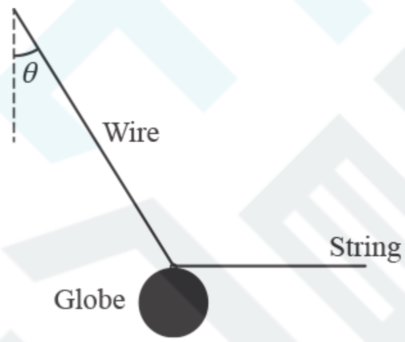


Fig. 7

Find the set of values of θ for which the tension in the string is less than the weight of the globe. **[4]**

19 Austin pushes a curling stone of mass 18 kg on ice. He releases the stone and it slides 21 m in a straight line until coming to rest after 14 s. He models the motion by assuming that the frictional force is constant.

(a) Show that the speed of the stone at the moment that Austin releases it is 3 m s^{-1}. **[2]**

(b) Find the magnitude of the frictional force acting on the stone. **[3]**

(c) Calculate the coefficient of friction between the stone and the ice, giving your answer correct to 2 significant figures. **[3]**

20 A car pulls a caravan along a straight road uphill on a slope that makes an angle of 8° with the horizontal. The mass of the car is 1700 kg and the mass of the caravan is 1300 kg. The driving force is 9000 N and the car and caravan experience resistances of 600 N and 1200 N respectively. The coupling between the car and the caravan is light and parallel to the road.

(a) Draw a diagram showing all the forces acting on the car and all the forces acting on the caravan. **[3]**

(b) Find the acceleration of the car and caravan as they move up the hill. **[4]**

(c) Find the tension in the coupling between the car and the caravan. **[3]**

21 Fig. 1 shows a block of mass 5 kg on a rough plane inclined at an angle α to the horizontal.

The block is in equilibrium.

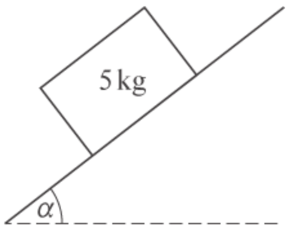


Fig. 1

(i) Draw a force diagram showing all the forces acting on the block. **[3]**

(ii) The normal reaction of the plane on the block is 37.5 N.

Find α, giving your answer to the nearest degree.

Find also the frictional force acting on the block. **[3]**

22 Olga and Petya are using light ropes to pull a sledge across rough snow.

- The surface of the snow is horizontal.
- The mass of the sledge and its load is 430 kg.
- Both ropes are horizontal.
- Olga pulls with a force of 120 N at an angle of 20° to the line of motion of the sledge.
- Petya also pulls with a force of 120 N at an angle of 20° to the line of motion of the sledge.

This is illustrated in a plan view in Fig. 2.

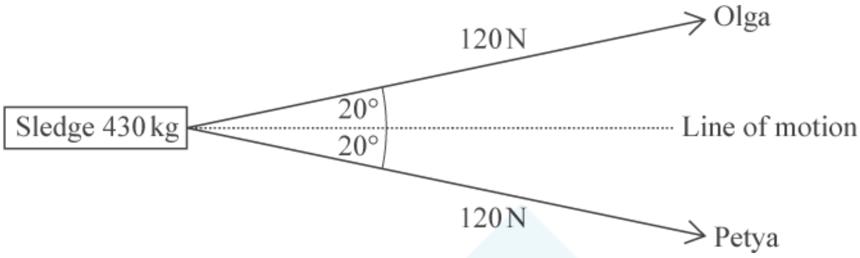


Fig. 2

(i) The sledge has acceleration 0.05 m s^{-2} in the direction of its line of motion.

Find the frictional force acting on the sledge. **[3]**

Olga and Petya then change to walking side by side. Their ropes, which are still horizontal, are now along the line of motion of the sledge. They maintain the forces on their ropes at 120 N and the frictional force remains the same.

(ii) Find the percentage increase in the acceleration of the sledge. **[4]**

23 A non-uniform rod 0.8 m long rests horizontally on smooth pegs A and B at each end of the rod. The contact forces at A and B are 10 N and 15 N respectively, as shown in Fig. 2.

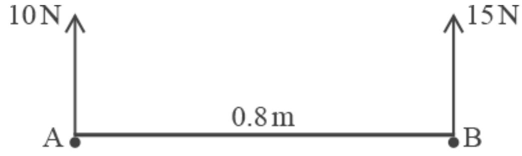


Fig. 2

Calculate the distance of the centre of mass of the rod from A. **[3]**

24 A block slides from rest down a line of greatest slope of a smooth plane inclined at 20° to the horizontal. Calculate the speed of the block when it has travelled 2 m. **[4]**

25 Fig. 9 shows a block of mass 2 kg resting on a rough horizontal table. It is attached to a ball of mass *m* kg by a light inextensible string that passes over a smooth pulley at the edge of the table. The ball hangs vertically below the pulley. The coefficient of friction between the block and the table is 0.6.

The system is held in equilibrium by a force of $5\sqrt{2}$ acting on the block at 45° above the horizontal. The block is on the point of sliding towards the pulley.

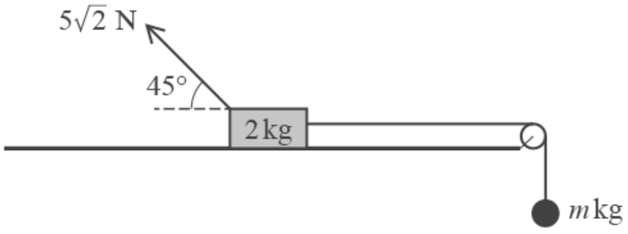


Fig. 9

(a) Complete the force diagram above to show all the forces acting on the block and the ball. **[2]**

(b) Calculate the frictional force acting on the block. **[4]**

(c) Calculate the value of m. **[4]**

26 In this question, **i** is a horizontal unit vector and **j** is a unit vector directed vertically upwards.

A particle is projected from the origin with an initial velocity of $(u_1\mathbf{i} + u_2\mathbf{j})\text{ms}^{-1}$, and moves freely under gravity. Its position vector **r** m at time t s is given by

$$\mathbf{r} = (u_1\mathbf{i} + u_2\mathbf{j})\ t - 5t^2\mathbf{j}.$$

(a) Write down the value of g used in this model. **[1]**

(b) Explain what is meant by the statement that g is not a universal constant. **[1]**

The position vector of the particle when it reaches its maximum height is (14**i** + 20**j**) m.

(c) Determine the initial velocity of the particle, giving your answer as a vector. **[7]**

(d) The particle hits a building which is 21 m away from the origin in the **i** direction. Calculate the height above the level of the origin at which the particle hits the building. **[3]**

27 Fig. 11 shows two blocks at rest, connected by a light inextensible string which passes over a smooth pulley. Block A of mass 4.7 kg rests on a smooth plane inclined at 60° to the horizontal. Block B of mass 4 kg rests on a rough plane inclined at 25° to the horizontal. On either side of the pulley, the string is parallel to a line of greatest slope of the plane. Block B is on the point of sliding up the plane.

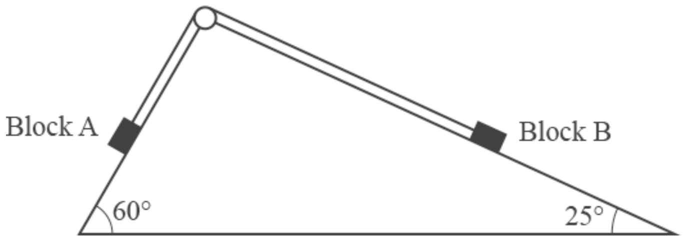


Fig.11

(a) Show that the tension in the string is 39.9 N correct to 3 significant figures. [2]

(b) Find the coefficient of friction between the rough plane and Block B. [5]

28 (a) (i) Sketch the graph of $y = 3^x$. **[1]**

(ii) Give the coordinates of any intercepts. **[1]**

The curve $y = f(x)$ is the reflection of the curve $y = 3^x$ in the line $y = x$.

(b) Find $f(x)$. **[1]**

29

(a) Express $7\cos x - 24\sin x$ in the form $R\cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. **[3]**

(b) Write down the range of the function

$$f(x) = 12 + 7\cos x - 24\sin x, \qquad 0 \le x \le 2\pi.$$

[2]

30 A particle is in equilibrium under the action of three forces in newtons given by

$$\mathbf{F}_1 = \begin{pmatrix} 8 \\ 0 \end{pmatrix}, \mathbf{F}_2 = \begin{pmatrix} 2a \\ -3a \end{pmatrix} \text{ and } \mathbf{F}_3 = \begin{pmatrix} 0 \\ b \end{pmatrix}.$$

Find the values of the constants *a* and *b*. **[3]**

31 Zoe tries to push a box of mass 5 kg along a rough horizontal floor. When she applies a horizontal force of P N the box is on the point of sliding. When she applies a horizontal force of $3P$ N the box has an acceleration of 2 m s^{-2}. Find the value of P. **[3]**

32 Fig. 7 shows a rectangular lamina ABCD with sides AB of length 60 cm and AD of length 50 cm. The lamina is lying flat on a smooth horizontal surface, and is acted on by the following five horizontal forces.

- 45 N at A in the direction BA
- 40 N at D in the direction AD
- 27 N at C in the direction BC
- *Y* N at A in the direction DA
- *X* N at E in a direction perpendicular to the edge BC

The lamina is in equilibrium. The distance BE is *d* cm.

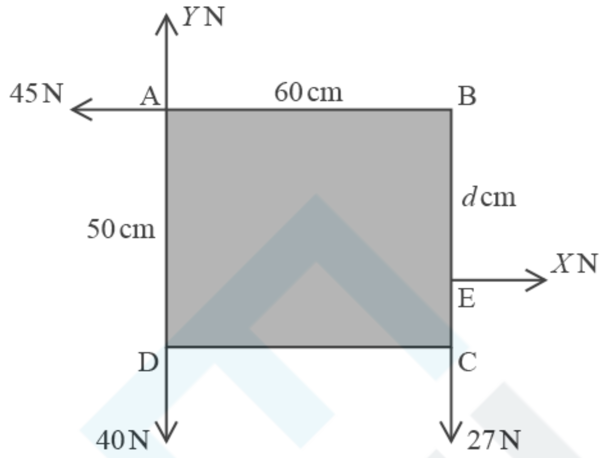


Fig. 7

Calculate the values of each of the following.

- *X*
- *Y*
- *d*

[4]

33 A block of mass 2.7 kg is placed on a rough plane inclined at 25° to the horizontal. The coefficient of friction between the block and the plane is 0.4. A force of 24 N parallel to a line of greatest slope of the plane pulls the block up the plane from rest.

(a) Calculate the acceleration of the block. **[5]**

After 5 s the 24 N force is removed.

(b) Calculate the distance that the block travels after the force is removed before coming instantaneously to rest. **[6]**

34 A block of mass 5 kg is placed on a rough horizontal table. The coefficient of friction between the table and the block is 0.3. A horizontal force P N is applied to the block but the block does not move.

Find the greatest possible value of P. **[4]**

35 Fig. 3 shows a rod AB which is 0.9 m long and hangs vertically from a smooth hinge at A. The rod can rotate about A in a vertical plane. Forces of 100 N and 60 N act at right angles to AB in this plane. Their points of application are 0.3 m and 0.75 m respectively below A.

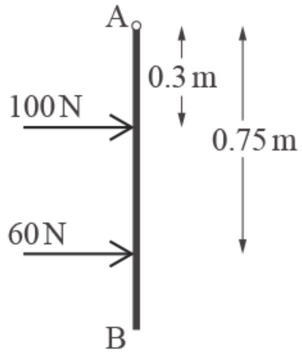


Fig. 3

(a) Find the combined moment of these forces about A. [2]

The rod is held in equilibrium by a force of F N which is also at right angles to the rod in the same vertical plane.

(b) Find the least possible value of F. [2]

(c) Explain how the rod can be in equilibrium when the resultant of these three forces is not zero. [2]

36 A model train consists of an engine of mass 0.3 kg and a truck of mass 0.2 kg. The train is pulled up a smooth plane inclined at 20° to the horizontal by a force of 2.5 N. This force is applied to the engine and acts parallel to a line of greatest slope of the plane.

(a) Show that the acceleration of the train is 1.65ms^{-2} correct to 3 significant figures. **[3]**

(b) Find the tension in the coupling between the engine and the truck. **[2]**

(c) Find the distance travelled by the train as it accelerates from 1ms^{-1} to 3ms^{-1}. **[2]**

37(a) In this question, the unit vectors **i** and **j** are horizontal and vertically upwards respectively.

A particle has mass 2.5 kg.

Write the weight of the particle as a vector. **[1]**

(b) The particle moves under the action of its weight and two external forces $(3\mathbf{i} - 2\mathbf{j})$ N and $(-\mathbf{i} + 18\mathbf{j})$ N.

Find the acceleration of the particle, giving your answer in vector form. **[2]**

38 Fig. 6 shows a train consisting of an engine of mass 80 tonnes pulling two trucks each of mass 25 tonnes.

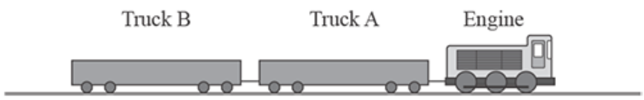


Fig. 6

The engine exerts a driving force of D N and experiences a resistance to motion of 2000 N. Each truck experiences a resistance of 600 N. The train travels in a straight line on a level track with an acceleration of 0.1 m s^{-2}.

Complete the force diagram provided below to show all the forces acting on the engine and each of the trucks. **[3]**

Force diagram

39(a) In this question, the unit vector **i** is horizontal and the unit vector **j** is vertically upwards.

A particle of mass 0.8 kg moves under the action of its weight and two forces given by (k**i** + 5**j**) N and (4**i** + 3**j**) N. The acceleration of the particle is vertically upwards.

Write down the value of k. **[1]**

(b) Initially the velocity of the particle is (4**i** + 7**j**) ms^{-1}.

Find the velocity of the particle 10 seconds later. **[4]**

40(a) A 15 kg box is suspended in the air by a rope which makes an angle of 30° with the vertical. The box is held in place by a string which is horizontal.

Draw a diagram showing the forces acting on the box. **[1]**

(b) Calculate the tension in the rope. **[2]**

(c) Calculate the tension in the string. **[2]**

41(a) A particle of mass 2 kg slides down a plane inclined at 20° to the horizontal. The particle has an initial velocity of 1.4 ms^{-1} down the plane. Two models for the particle's motion are proposed.

In model A the plane is taken to be smooth.

Calculate the time that model A predicts for the particle to slide the first 0.7 m. **[5]**

(b) Explain why model A is likely to underestimate the time taken. **[1]**

(c) In model B the plane is taken to be rough, with a constant coefficient of friction between the particle and the plane.

Calculate the acceleration of the particle predicted by model B given that it takes 0.8 s to slide the first 0.7 m. **[2]**

(d) Find the coefficient of friction predicted by model B, giving your answer correct to 3 significant figures. **[6]**

42(a) The function $f(x)$ is defined for all real x by

$f(x) = 3x - 2.$

Find an expression for $f^{-1}(x)$. [2]

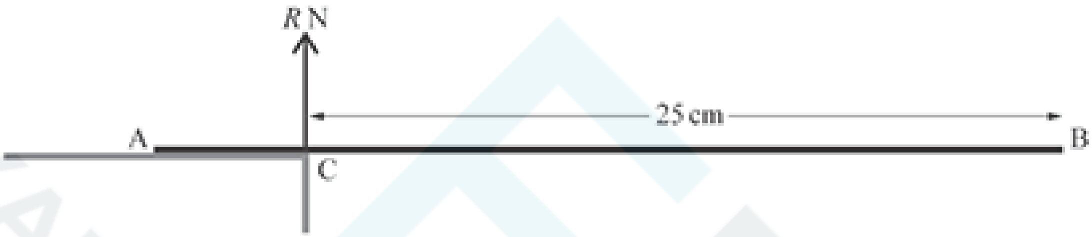
(b) Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram. [2]

43(a) A uniform ruler AB has mass 28 g and length 30 cm. As shown in Fig. 6, the ruler is placed on a horizontal table so that it overhangs a point C at the edge of the table by 25 cm.

A downward force of F N is applied at A. This force just holds the ruler in equilibrium so that the contact force between the table and the ruler acts through C.

Fig. 6

Complete the force diagram below, labelling the forces and all relevant distances. **[2]**



(b) Calculate the value of F. **[2]**

44(a) A block of mass 2 kg is placed on a rough horizontal table. A light inextensible string attached to the block passes over a smooth pulley attached to the edge of the table. The other end of the string is attached to a sphere of mass 0.8 kg which hangs freely.

The part of the string between the block and the pulley is horizontal. The coefficient of friction between the table and the block is 0.35. The system is released from rest.

Draw a force diagram showing all the forces on the block and the sphere. **[3]**

(b) Write down the equations of motion for the block and the sphere. **[2]**

(c) Show that the acceleration of the system is 0.35 m s^{-2}. **[4]**

(d) Calculate the time for the block to slide the first 0.5 m. Assume the block does not reach the pulley. **[2]**

45(a) Fig. 15 shows a particle of mass m kg on a smooth plane inclined at 30° to the horizontal. Unit vectors **i** and **j** are parallel and perpendicular to the plane, in the directions shown.

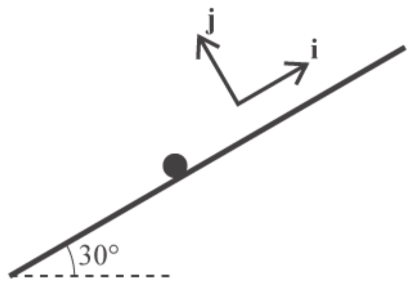


Fig. 15

Express the weight **W** of the particle in terms of m, g, **i** and **j**. **[2]**

(b) The particle is held in equilibrium by a force **F**, and the normal reaction of the plane on the particle is denoted by **R**. The units for both **F** and **R** are newtons.

Write down an equation relating **W**, **R** and **F**. **[1]**

(c) Given that **F** = 6**i** + 8**j**,

• show that m = 1.22 correct to 3 significant figures,

• find the magnitude of **R**. **[6]**

46(a) In this question, the *x*, and y directions are horizontal and vertically upwards respectively.

A particle of mass 1.5 kg is in equilibrium under the action of its weight and forces $\mathbf{F}_1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \mathrm{N}$ and **F** $_2$.

Find the force **F** $_2$. **[3]**

(b)

The force **F** $_2$ is changed to $\begin{pmatrix} 2 \\ 20 \end{pmatrix} \mathrm{N}$.

Find the acceleration of the particle. **[3]**

47(a) Functions f(x) and g(x) are defined as follows.

$f(x) = \sqrt{x}$ for $x > 0$ and $g(x) = x^3 - x - 6$ for $x > 2$.

The function h(x) is defined as

$h(x) = fg(x)$.

Find h(x) in terms of x and state its domain. **[2]**

(b) Find h(3). **[1]**

(c) Fig. 15 shows $h(x)$ and $h^{-1}(x)$, together with the straight line $y = x$.

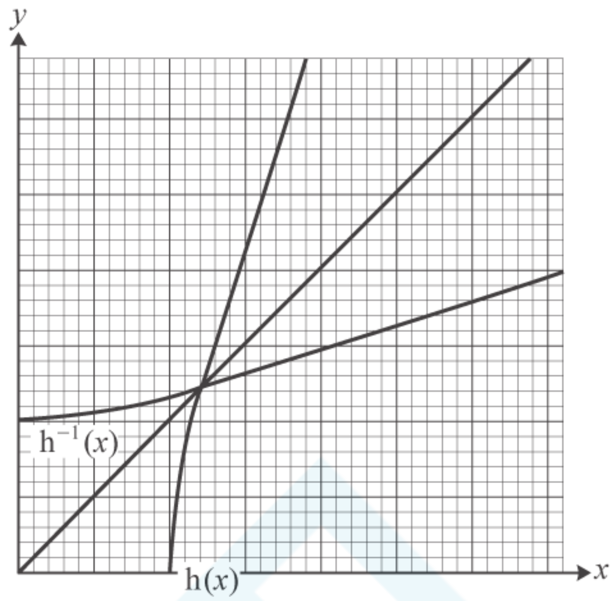


Fig. 15

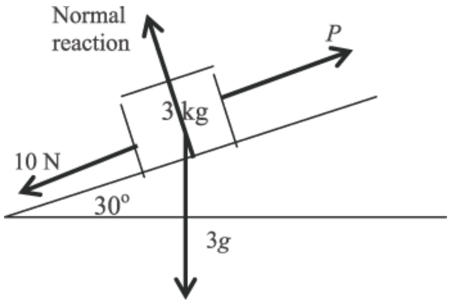
Determine the gradient of $y = h^{-1}(x)$ at the point where $y = 3$. **[4]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1		i	*(A)* The pulley is smooth	B1	Award for "smooth" seen.
		i	*(B)* Horizontal equilibrium: $T \sin \theta = T \sin \varphi$	M1	Attempt at horizontal equilibrium. Allow sin-cos interchange. The argument must be based on forces.
		i	$\Rightarrow \theta = \varphi$	E1	Do not allow if sin-cos interchange
		ii	Call M the mid point of AB. AM = 1, AC=1.4, $\angle$AMC = 90°	M1	Setting up triangle and use of trigonometry
		ii	Pythagoras $\Rightarrow$ $\mathbf{MC} = \sqrt{1.4^2 - 1^2} = \sqrt{0.96}$		
		ii	$\cos\theta = \frac{\sqrt{0.96}}{1.4} = \frac{\sqrt{24}}{7}$	E1	If decimals are matched, at least 3 figures must be given
		iii	Vertical equilibrium	M1	Use of vertical equilibrium
		iii	$2T \cos \theta = 50$	A1	Accept $T \cos \theta = 25$ as an equivalent statement
		iii	$T = 35.7$N	A1	Cao
		iv	$1.2^2 + 1.6^2 = 2^2$ $\Rightarrow \angle$ ACB = 90°	B1	Use of Pythagoras, or equivalent
		iv	$\cos \alpha = 0.6$, $\cos \beta = 0.8$	B1	Both No marks for sin-cos interchange
		v	**Either resolving horizontally and vertically**	5	
		v	$T_1 \cos \alpha = T_2 \cos \beta$	M1	Attempt at horizontal equation. Allow consistent sin-cos interchange
		v	$T_1 \sin \alpha + T_2 \sin \beta = 50$	M1	Attempt at vertical equation. Allow consistent sin-cos interchange
		v	$0.6T_1 = 0.8T_2$ $0.8T_1 + 0.6T_2 = 50$	A1	Substitution in both equations. Dependent on both M marks. Cao
		v	Solving simultaneously	M1	Dependent on both the previous M marks
		v	$T_1 = 40$, $T_2 = 30$	A1	Cao
		v	**Or resolving in the direction of the strings**		
		v	Resolving in both directions	M1	A serious attempt to use this method. Allow sin-cos interchange
		v	$T_1 = 50 \sin \alpha$	M1	
		v	$\Rightarrow T_1 = 50 \times 0.8 = 40$	A1	

Question			Answer/Indicative content	Marks	Guidance
		v	$T_2 = 50 \times \sin \beta$	M1	
		v	$\Rightarrow T_2 = 50 \times 0.6 = 30$	A1	
		v	Or triangle of forces		
		v	Use of a triangle of forces	M1	The triangle must be closed and have a right angle opposite the weight
		v	Labels	M1	The sides must be correctly annotated
		v	Angles	M1	The angles must be correctly annotated
		v	$T_1 = 50 \times 0.8 = 40$	A1	Cao Dependent of first M mark
		v	$T_2 = 50 \times 0.6 = 30$	A1	Cao Dependent of first M mark

Question			Answer/Indicative content	Marks	Guidance
		vi	Attempt to find ∠CAB	M1	May be implied by the remaining answers **Examiner's Comments** The final question was about different ways of suspending a block from two fixed points on a beam using a string of a given length. In the first three parts it was done using a smooth pulley. Initially candidates were expected to use mechanics and geometry to establish the symmetry of the situation; many lost marks on this but they were frequently able to recover to find the tension in the string correctly in part (iii). Parts (iv) and (v) dealt with a different situation in which the string was cut into two unequal parts, and this proved rather more familiar to candidates. In part (v), they were asked to find the tensions in the strings and, although most knew what they were trying to do, there were many careless mistakes with cos-sin interchanges and sign errors common. Most candidates resolved in the horizontal and vertical directions and tried to solve the resulting simultaneous equations; however, some used one of two other available methods (resolving in the directions of the strings, and a triangle of forces); because the strings were at right angles either of these required less work. In the last part of the question the situation was considered in which the string was cut at a different point; only a few candidates saw that with the given lengths all the weight would be carried by one of the strings and the other would be slack.
		vi	Tension in AC is 50 N (it takes all the weight)	B1	
		vi	Tension in BC is zero (it is slack)	B1	
			Total	18	

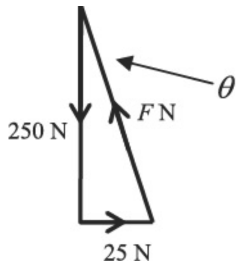
Question			Answer/Indicative content	Marks	Guidance
2		i		B1	
		i		B1	3 marks –1 / error or omission
		i		B1	Forces must have arrows and labels Accept "weight" and "friction"
		ii	$R = 3\,g\cos 30° = 25.46\ldots = 25.5$ (to 3 significant figures)	B1	Accept 25 or 26
		iii	$P = 10 + 3g\sin 30^{\circ}$	M1	Correct elements must be present
		iii	$P = 24.7$	A1	Cao Examiner's Comments This question, about a block moving on a slope, was well answered. Most candidates scored full marks on the force diagram in part (i), an improvement on similar questions in previous papers. In part (ii) a few candidates did not resolve the weight correctly, or in some cases at all, to find the normal reaction. Part (iii) considered forces parallel to the slope and this was well answered.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
3		i	$v = u + at$	M1	Use of a suitable constant acceleration formula
		i	$5 = 0 + a \times 10 \Rightarrow a = 0.5$	A1	Notice The value of *a* is not required by the question so may be implied by subsequent working
		i	$F = ma \Rightarrow 120 - R = 40 \times 0.5$	M1	Use of Newton's 2nd Law with correct elements
		i	$R = 100\text{N}$	E1	**Examiner's Comments** This question involved a sledge being pulled, initially horizontally and then up a slope. Part (i) asked for the resistance to motion and required the use of a constant acceleration formula and then Newton's 2nd Law. It was very well answered. A few candidates lost marks by using the given final answer in an argument that was less than a valid verification.
		ii	$(A)\ F = ma \Rightarrow -100 = 40a$	M1	Equation to find *a* using Newton's 2nd Law
		ii	$\Rightarrow a = -2.5$	A1	
		ii	When $t = 1.6$ $v = 5 + (-2.5) \times 1.6 = 1\ \text{ms}^{-1}$	A1	CAO **Examiner's Comments** This question involved a sledge being pulled, initially horizontally and then up a slope. The situation changed because the rope pulling the sledge broke. Candidates were asked to find the speed of the sledge at a time when it was still moving.

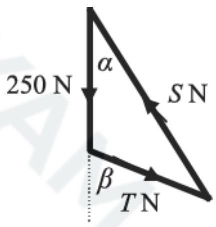
Question			Answer/Indicative content	Marks	Guidance
		ii	(B) When $t = 6$, it is stationary. $v = 0$ ms^{-1}	B1	Examiner's Comments□□ In part (ii) the situation changed because the rope pulling the sledge broke. In part (A) candidates were asked to find the speed of the sledge at a time when it was still moving and in part (B) at a later time when it would have come to a halt. Most candidates obtained the right answers to both parts. However, a few did not recognise that the acceleration changed when the rope broke and continued with the same value as they had in part (i). A more common mistake was to give a negative speed in part (B) rather than zero.
		iii	Motion parallel to the slope:	B1	Component of the weight down the slope, ie $40g\sin 15°$ (= 101.457…)
		iii	$200 - 40g\sin 15° = 40a$	M1	Equation of motion with the correct elements present. No extra forces.
		iii	$a = 2.463\ldots$		This result is not asked for in the question
		iii	$v^2 - u^2 = 2as \Rightarrow 8^2 = 2 \times 2.46\ldots \times s$	M1	Use of a suitable constant acceleration formula, or combination of formulae. Dependent on previous M1.
		iii	$\Rightarrow s = 12.989\ldots$ rounding to 13.0 m	E1	Note If the rounding is not shown for s the acceleration must satisfy $2.452\ldots < a < 2.471\ldots$ Examiner's Comments In part (iii) the sledge was being pulled up a smooth slope. There were many correct answers to this part but a few candidates were unable to use the component of the weight down the slope.
		iv	Let a be acceleration up the slope		
		iv	$-40 \times 9.8 \times \sin 15° = 40a$	M1	Use of Newton's 2nd Law parallel to the slope
		iv	$a = -2.536\ldots$, ie 2.536 ms^{-2} down the slope	A1	Condone sign error
		iv	$s = ut + \frac{1}{2}at^2$		

Question			Answer/Indicative content	Marks	Guidance
		iv	$-12.989... = 8t + \frac{1}{2}\times(-2.536...)t^2$	M1	Dependent on previous M1. Use of a suitable constant acceleration formula (or combination of formulae) in a relevant manner.
		iv	$1.268...t^2 - 8t - 12.989... = 0$	A1	Signs must be correct
		iv	$t = \dfrac{8 \pm \sqrt{64 - 4\times1.268...\times(-12.989...)}}{2\times1.268...}$	M1	Attempt to solve a relevant three-term quadratic equation
		iv	t = –1.339… or 7.647… , so 7.65 seconds	A1	
		iv	Alternative 2-stage motion		
		iv	Let *a* be acceleration up the slope		
		iv	–40 × 9.8 × sin15° = 40*a*	M1	Use of Newton's 2nd Law parallel to the slope
		iv	*a* = –2.536… , ie 2.536 ms^{-2} down the slope	A1	Condone sign error
		iv	Motion to highest point		
		iv	$v = u + at \Rightarrow 0 = 8 - 2.536...t$	M1	Dependent on previous M1. Use of a suitable constant acceleration formula, for either *t* or *s*, in a relevant manner.
		iv	*t* = 3.154…	A1	For either *t* or *s*
		iv	$s = ut + \frac{1}{2}at^2 \Rightarrow s = 8\times3.154... - \frac{1}{2}\times2.536...\times3.154...^2$		
		iv	*s* = 12.616…		
		iv	Distance to bottom = 12.989… + 12. 616… = 25.605…		
		iv	$s = ut + \frac{1}{2}at^2 \Rightarrow 25.605... = \frac{1}{2}\times2.536...\times t^2$	M1	Use of a suitable constant acceleration formula
		iv	*t* = 4.493…		

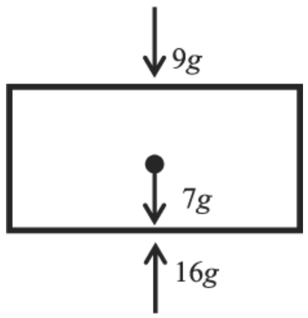
Question			Answer/Indicative content	Marks	Guidance
		iv	Total time = 3.154… + 4.493… = 7.647… s	A1	Examiner's Comments□□ In part (iv), there was no longer a pulling force (the rope had broken again) and the sledge started moving up the slope, came to a stop and then slid down to the bottom of the slope. Candidates were asked to find how long this took. Many candidates knew what they had to do and there were plenty of correct answers; however, there were also many sign errors. This was the last question on the paper and several low-scoring candidates did not get started on it. There were also those who substituted completely wrong numbers into their constant acceleration formulae, indicating incorrect analysis of the situation.
			Total	18	

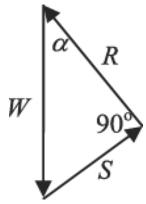
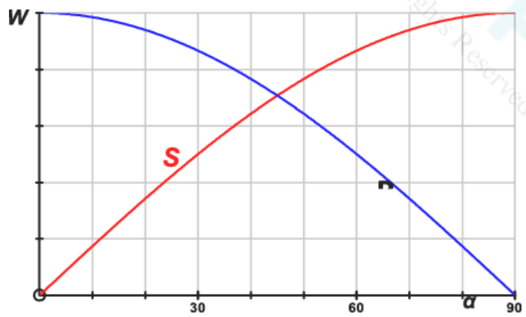
Question			Answer/Indicative content	Marks	Guidance
4		i		B1 B1 B1	Shape of triangle; ignore position of θ if marked in diagram 2 marks –1 per error but penalise no arrows only once and penalise no labels only once. Condone *T* written for *F*. In the case of a force diagram showing F, 25 and 250 allow maximum of 2 marks with –1 per error but penalise no arrows only once and penalise no labels only once
		i			**Examiner's Comments** This question was about forces in equilibrium. It was set in the context of two people hoisting an object towards the top of a building. In part (i), candidates were asked to draw a triangle of forces. While there were plenty of correct answers many marks were lost through incorrect or missing labelling and the absence of arrows. A lot of candidates drew a force diagram instead and so could only obtain 2 out of the 3 marks.
		ii	$\tan\alpha = \dfrac{25}{250}$	M1	M1 for recognising and using α in the triangle
		ii	⇒ α = 5.7°	A1	
		ii	$F = \sqrt{25^2 + 250^2}$	M1	Use of Pythagoras
		ii	F = 251.2	A1	At least 3 significant figures required
		ii	Distance = 30 tanα = 30 × 0.1 = 3m	B1	CAO
		ii	Alternative F cosθ = 250 F sinθ = 25		
		ii	$\tan\theta = \dfrac{25}{250}$	M1	
		ii	⇒ θ = 5.7°	A1	
		ii	F cos5.7° = 250	M1	
		ii	F = 251.2	A1	At least 3 significant figures required
		ii	Distance = 30tanα = 30 ☐ 0.1 = 3 m	B1	CAO

Question			Answer/Indicative content	Marks	Guidance
		ii			**Examiner's Comments** Candidates who had drawn a correct triangle of forces in part (i) were usually successful in part (ii), which asked for information that could easily be obtained from it. Those who had drawn force diagrams in part (i) could still answer part (ii) and many did so successfully but usually after a little more work.
		iii	Vertical equilibrium	M1	M1 for attempt at resolution in an equation involving both S and T; condone sin-cos errors for the M mark only
		iii	↑ $S\cos\alpha = T\cos\beta + 250$ ↓	A1	
		iii	Horizontal equilibrium $S\sin\alpha = T\sin\beta$	A1	
		iii			**Examiner's Comments** In part (iii) the situation had changed and many of those candidates who had made mistakes in the earlier parts were able to recover. The question asked for the vertical and horizontal equilibrium equations and there were many correct answers. Common errors involved incorrect signs or the omission of one of the forces.
		iv	$S\sin 8.5° = T\sin 35° \Rightarrow S = 3.8805T$	M1	Using one equation to make S or T the subject in terms of the other
		iv	$(3.8805T)\cos 8.5° = T\cos 35° + 250$	M1	Substituting in the other equation
		iv	$T = 82.8$	A1	CAO
		iv	$S = 321.4$	A1	CAO
		iv	**Alternative**		**Use of linear simultaneous equations**
		iv	$S\sin 8.5° - T\sin 35° = 0$		
		iv	$S\cos 8.5° - T\cos 35° = 250$		
		iv	$S\sin 8.5°\cos 35° - T\sin 35°\cos 35° = 0$		
		iv	$S\cos 8.5°\sin 35° - T\cos 35°\sin 35° = 250\sin 35°$		

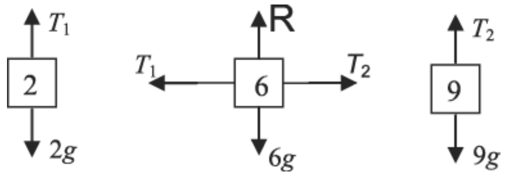
Question	Answer/Indicative content	Marks	Guidance
iv	S (–sin8.5°cos35° + cos8.5°sin35°) = 250sin35°	M1	Valid method that has eliminated terms in either S or T (execution need not be perfect)
iv	S = 321.4	A1	CAO First answer
iv	Substituting in either equation	M1	Substituting to find the second answer
iv	⇒ T = 82.8	A1	CAO Second answer
iv	Alternative Triangle of forces		
			Either Drawing and using a triangle of forces
iv		M1	Or Quoting and using Lami's Theorem
iv	$\frac{S}{\sin 145°} = \frac{T}{\sin 8.5°} = \frac{250}{\sin 26.5°}$	M1	Correct form of these equations
iv	S = 321.4	A1	CAO
iv	T = 82.8	A1	CAO
			<u>Examiner's Comments</u>
			In part (iv) candidates were asked to solve the equations they had obtained in part (iii) with particular values given for the two angles. There were many right answers but also many careless mistakes.
v	Abi's weight is 40g = 392 N	M1	Consideration of Abi's weight
v	When α = 60°, S cos60° > 250 ⇒ S > 500	M1	Consideration of vertical forces on the object. Condone no mention of Bob's rope
v	The tension in rope A would be greater than Abi's weight and so she would be lifted off the ground	A1	The argument must be of high quality and must include consideration of the tension in Bob's rope
v	**Alternative**		

Question			Answer/Indicative content	Marks	Guidance
		v	If Abi is on the ground, the maximum possible tension in rope A is Abi's weight of 392 N	M1	Consideration of Abi's weight
		v	So the maximum upward force on the object is 392 × cos60° = 192 N		
		v	This is less than the weight of the object, and the tension in Bob's rope is pulling the box down.	M1	Consideration of vertical forces on the object. Condone no mention of Bob's rope Or the box accelerated downwards
		v	So Abi would be lifted off the ground	A1	The argument must be of high quality and must include consideration of the tension in Bob's rope __Examiner's Comments__ In part (v) candidates were presented with another situation and asked to explain why it was impossible. This was probably the most challenging question on the paper. There were a few excellent answers but many candidates did not present a coherent argument.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
5				 B1 B1 B1	One mark for each force with correct magnitude and direction Deduct 1 mark only for *g* missing 16g ↑ 7g ↓ 9g ↓ If all three forces are correct but there is at least one extra force, deduct 1 mark and so give 2 marks. Otherwise ignore extra forces. Note For 16g ↑ 16g ↓ Award B1 B0 B0 Examiner's Comments☐ This question, about drawing a force diagram, was not well answered. Candidates were expected to identify the three forces acting on a block and to mark each of them on a given diagram. Many tried to combine two of them, even though they were quite different forces; other answers can only be described as chaotic.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
6		i			Closed triangle with cycling arrows. Accept any consistent orientation. All forces labelled. Correct angles. The 90° may be implied. α may be shown between *S* and the horizontal (ie outside the triangle). SC1 Award for a force diagram with no extra forces and all labels and directions correct.
		i		B1	Closed triangle with cycling arrows. Accept any consistent orientation.
		i		B1	All forces labelled.
		i		B1	Correct angles. The 90° may be implied. α may be shown between *S* and the horizontal (ie outside the triangle). SC1 Award for a force diagram with no extra forces and all labels and directions correct.
		ii	$R = W\cos\alpha$	B1	Allow FT for sin-cos interchange following the wrong angle in the triangle being marked α in part (i) for both marks.
		ii	$S = W\sin\alpha$	B1	SC1 if both *S* and *R* are given negative signs
		iii			The answers in part (iii) must - either be fully correct - or they must all be consistent with those in part (ii) where the marks in part (ii) are FT from part (i). No credit should be given to forms other than *W*cosα and *W*sinα. The curves must have the correct end points and lie withinthe correct range; no credit should be given for straight lines. Graphs must be correctly labelled. Unlabelled graphs get B0 B0.

Question			Answer/Indicative content	Marks	Guidance
		iii	Sketch graph of R against α	B1	Condone no explicit vertical scale. Do not accept straight lines.
		iii	Correct sketch graph of S against α	B1	Must be consistent with graph of R
		iii	$45° < \alpha$ ($\leq 90°$)	B1	Condone $45° \leq \alpha$ **Examiner's Comments** This was a statics question involving three forces. In the first part candidates were asked to draw a triangle of forces and many dropped some of the marks for this; lack of arrows, incorrect or missing labels and errors in the directions were all quite common. A number of candidates drew a force diagram instead and at most only one of the three marks was available for this. The final part of the question required candidates to draw graphs of Wsin and Wcos; many candidates found this difficult and a significant number made no attempt at an answer.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
7		i		B1	Diagrams for both 2 and 9 kg blocks. The tensions must be different from each other. No extra forces.
		i		B1	Tensions on 6 kg block. The tensions must be different from each other. No extra forces.
		i		B1	$6g$ and R on 6 kg block. No extra forces. **Special Case** When the tensions are given as T_1, T_2, T_3, T_4 (or equivalent) award up to SC1 SC0 for the first two marks. **Examiner's Comments** Candidates were asked to draw separate diagrams showing the forces on the three objects. There were two strings and the tensions in them were different. However, many candidates treated them as the same and just marked them as *T*. This was obviously a serious mistake which undermined the whole question and so caused a large loss of marks both in part (i) and, if continued.
		ii	$9g - T_2 = 9a$	M1	First equation correct
		ii	$T_2 - T_1 = 6a$	M1	Both the remaining two equations correct.
		ii	$T_1 - 2g = 2a$		Do not give this mark if both tensions are shown as the same.
		ii	$a = \frac{7}{17}g = 4.04 \text{ (m s}^{-2}\text{)}$	A1	The final three marks are dependent on both M marks a, T_1 and T_2 may be found in any order and FT should be allowed from the first of these found
		ii	$T_1 = 27.7$ (N)	A1	
		ii	$T_2 = 51.9$ (N)	A1	
		ii	**Alternative: Whole system**		
		ii	$9g - 2g = 17a$	M1	

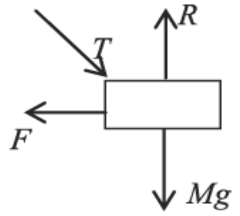
Question			Answer/Indicative content	Marks	Guidance
		ii	$a = \frac{7g}{17} = 4.04$	A1	
		ii	$T_1 - 2g = 2a$ and $9g - T_2 = 9a$	M1	Both equations correct. Oe.
		ii	$T_1 = 27.7$ (N)	A1	The final two marks are dependent on both M marks. T_1 and T_2 may be found in either order and FT should be allowed from their value for a.
		ii	$T_2 = 51.9$ (N)	A1	**Examiner's Comments** Candidates went on to find the acceleration of the system and the tensions in the two strings. Those who had drawn correct diagrams in part (i) usually got this completely correct but those who had not done so, rarely made much progress on this part.
			Total	8	
8			$P = 8\sqrt{2}\sin 45° + 12\sin 30°$	M1 M1	Considering equilibrium in the vertical direction Resolution of forces of 12 N and $8\sqrt{2}$ N in the vertical direction. Do not allow sin-cos interchange for the 30° angle.
			$P = 14$	A1	Dependent on both M marks
			$Q + 8\sqrt{2}\cos 45° = 12\cos 30°$	B1	
			$Q = 2.39$	B1	**Examiner's Comments** This question was answered correctly by almost all candidates. A small number made sign errors, particularly when finding Q. There were also those who did not give Q to 3 significant figures, as requested in the question.
			Total	5	

Question			Answer/Indicative content	Marks	Guidance
9		i	$\cos\alpha = 0.8$, $\cos\beta = 0.6$.	B1	Or equivalent statements **Examiner's Comments** This was the second of the long questions on the paper, worth 18 marks. It was set in the context of raising an unexploded bomb from a hole on a building site. This question was quite challenging and many candidates were unsuccessful on the later parts. The question started with a straightforward piece of trigonometry for 1 mark, and almost all candidates were successful. Answers 0.8 and 0.6.
		ii	Horizontal forces → $8000\cos\beta - 6000\cos\alpha$	M1	Do not allow sin-cos interchange
		ii	$4800 - 4800 = 0$ So the horizontal component of acceleration is 0	A1	Must state acceleration is zero
		ii	Vertical forces ↑ $T\sin\alpha + 8000\sin\beta - 7500$	M1	Do not allow if the weight is missing Allow $T\cos\beta + 8000\cos\alpha - 7500$
		ii	$\sin\alpha$ (= $\cos\beta$) =0.6 and $\sin\beta$ (= $\cos\alpha$) = 0.8	B1	o.e. CAO May be seen or implied in the working
		ii	$6400 + 3600 - 7500 = 2500$	A1	CAO
		ii	Mass of bomb $\frac{7500}{9.8}$ (=765.3) kg	M1	

Question			Answer/Indicative content	Marks	Guidance
		ii	$\alpha = \dfrac{2500}{765.3} = 3.27$ The acceleration is 3.27 ms^{-1} upwards	A1	CAO Allow 3.26 **Examiner's Comments** The question then went on to consider the horizontal and vertical components of acceleration in a particular situation. The first demand was to show that the horizontal component is zero. Most candidates got this right but some lost a mark by failing to take the step of going from zero resultant force to zero acceleration; this was a given result and so a high standard of argument was expected. The question then went on to find the vertical component of acceleration and this elicited many good answers. A few candidates failed to convert the weight of the bomb to its mass, and some missed it out completely. There were fewer sin-cos interchanges than might have been the case a few years ago. Answer 3.27 ms^{-2}.
		iii	No horizontal acceleration ☐ Resultant = 0		
		iii	Horizontal forces → $8000\cos\beta - T\cos\alpha = 0$	M1	Horizontal must be indicated
		iii	$T = \dfrac{8000\cos\beta}{\cos\alpha}$	A1	
		iii	$\cos\alpha = \dfrac{16}{\sqrt{d^2+16^2}}, \quad \cos\beta = \dfrac{9}{\sqrt{d^2+9^2}}$	M1	

Question			Answer/Indicative content	Marks	Guidance
		iii	$T = 8000 \times \dfrac{\dfrac{9}{\sqrt{d^2+81}}}{\dfrac{16}{\sqrt{d^2+256}}} = \dfrac{4500\sqrt{d^2+256}}{\sqrt{d^2+81}}$	A1	**Examiner's Comments** The question then went on to consider a general situation during the lift. The first request was derive a given result for *T*. Many candidates lost marks here by not relating it to the horizontal direction. Some candidates may not have been aware that because this was a given result a high standard of argument was expected. Candidates were then asked to show that the given result for *T* could be written in a different form. While there were plenty of correct answers, there were also many that appeared to conjure the given result out of incorrect working.
		iv	When $d = 6.75$, $T = 4500 \times \dfrac{\sqrt{6.75^2+256}}{\sqrt{6.75^2+81}}$ (= 6946.2 ...)	B1	May be implied by subsequent working Note In this situation $\alpha = 22.9°$, $\beta = 36.9°$
		iv	Vertical forces ↑ $6946.2\sin\alpha + 8000\sin\beta - 7500$	M1	Their α and β. No sin-cos interchange. Note The forces are 2700 N and 4800N
		iv	= 0	A1	Condone any resultant force that rounds to 0 to the nearest integer.
		iv	So the (vertical) acceleration is zero.	A1	CAO **Examiner's Comments** In this part of the question, the bomb was at the height at which its vertical acceleration was zero and candidates were expected to use the result given at the end of part (iii) to discover this. Only the stronger candidates were successful. Many of those who attempted to find the equation of motion used the wrong angles or the wrong tensions, or forgot about the weight completely. Answer Acceleration = 0 ms^{-2}
		iv	**Alternative** Vertical forces ↑ $T\sin\alpha + 8000\sin\beta - 7500$		
		iv	$4500 \times \dfrac{\sqrt{6.75^2+256}}{\sqrt{6.75^2+81}} \times \dfrac{6.75}{\sqrt{6.75^2+256}} + 8000 \times \dfrac{6.75}{\sqrt{6.75^2+81}} - 7500$	M1	

Question			Answer/Indicative content	Marks	Guidance
		iv		B1	
		iv	$12500 \times \frac{6.75}{11.25} - 7500 = 0$	A1	
		iv	So the (vertical) acceleration is zero.	A1	CAO
		v	When at P there would be no vertical components of the tensions to counteract the weight.	B1	
		v	The acceleration would be *g* vertically downwards.	B1	The acceleration must be stated to be *g* **Examiner's Comments** In part (ii) the bomb was in a position where it was accelerating upwards. In part (iv) it was in a position where equilibrium was possible but there could be no upwards acceleration. The final part of the question considered the hypothetical situation where the bomb was at the top and so was at the same level as the winches. Candidates were asked to explain why equilibrium was impossible in this situation and to state the acceleration. While there were some excellent explanations many were garbled or wrong. Many candidates said there would be zero acceleration. Answer g ms^{-2} vertically downwards.
			Total	18	

Question			Answer/Indicative content	Marks	Guidance
10		i		B1	Forces — **B0** if one force missing or an extra force present
		i		B1	Labels
		i		B1	Arrows — **B0** if T in tension Allow T given in components provided it is clear they are not additional forces. Allow sin-cos interchange in this case. Give **B0 B0 B0** if 2 or more forces missing **Examiner's Comments** In this question candidates were asked to draw a force diagram and many did this successfully. The most common mistake was to reverse the direction the thrust applied to the block, making it into a tension instead; a few candidates omitted the normal reaction. Some candidates replaced the thrust by its vertical and horizontal components and that was entirely acceptable provided that they were not presented as extra forces in addition to the actual thrust.
		ii			Notice that the same solution applies if the direction of T was wrong in part (i), and full marks are available for part (ii) in this case.
		ii	$T \cos \alpha - F = ma$	M1	Horizontal equation of motion with the right 3 elements
		ii	$40 \cos \alpha - F = 5 \times 1.5$	A1	A0 if sin-cos interchange

Question			Answer/Indicative content	Marks	Guidance	
		ii	F = 12.5 Frictional force of 12.5 N.	A1	CAO **Examiner's Comments** In part (ii) candidates were expected to apply Newton's 2nd law to the block and to deduce the frictional force acting on it. Most candidates got this right but there were a few sign errors. Answer 12.5 N	
			Total	6		
11		a	E.g. The particle is in equilibrium [and the given forces cannot sum to zero as at 90°]	B1(AO2.2a) [1]	oe	Accept “without another force present, the particle would be moving on a rough surface without a frictional force”
		b	Friction 10 N [to give horizontal resultant of 0] Normal reaction from table. 8 N [to give vertical resultant of 0] **Alternative method** One extra force that gives equilibrium. Components 10 N → and 8 N ↑ Components from Friction → and normal reaction ↑	B1(AO3.3) B1(AO1.2) B1(AO3.3) B1(AO1.2) [2]	oe Accept ‘Because the surface is rough’ for ‘Friction’ Oe oe Accept $\sqrt{164}$ at ≈ 39° to horizontal oe Accept ‘because the surface is rough’ for ‘Friction’	
			Total	3		

Question			Answer/Indicative content	Marks	Guidance
12		a	No acceleration so we require the weight $15g$ (147 N)	B1(AO3.3) [1]	Accept $15g$, $15g$ N, 147 N etc
		b	Tension T N, N2L $\uparrow$ $T - 147 = 15 \times 2$ So $T = 177$ and tension is 177 N	M1(AO3.4) A1(AO1.1) [2]	Application of N2L
		c	177 N	B1(AO3.4) [1]	FT from (b)

Question			Answer/Indicative content	Marks	Guidance
		d	Let the air resistance be $R\uparrow$	M1(AO3.3)	
			N2L $\uparrow$ gives $R + 140 - 15g = 15 \times (-1.5)$	M1(AO1.1)	Finding R using a and then T
			$R - 7 = -22.5$		
			$R = -15.5$	A1(AO1.1)	
			Hence magnitude is 15.5 N	A1(AO3.4)	
			R is downwards so motion of the package is upwards	E1(AO3.4)	
				[5]	
			Alternative method		
			Let the total upward force be F N and the air resistance $R\uparrow$		
			N2L $\uparrow$ gives $F - 147 = 15 \times (-1.5)$	M1(AO3.3)	Finding F and hence R using a and then T
			So $F = 124.5$	A1(AO1.1)	
			Also $F = R + 140$ so $R = 124.5 - 140 = -15.5$	A1(AO1.1)	
			Hence magnitude is 15.5 N	A1(AO3.4)	
			R is downwards so motion of the package is upwards	A1(AO3.4)	
				[5]	
			Total	9	

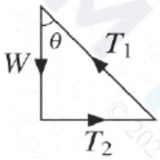
Question			Answer/Indicative content	Marks	Guidance
13		a	Horizontal component = 100 × cos32° 84.8 N	M1(AO3.3) A1(AO1.1) [2]	
		b	Frictional force is 100 cos θ [with $\theta < 32°$] So frictional force increases. (oe)	M1(AO3.4) E1(AO2.2a) [2]	
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
14		a	*A* Let acceleration be *a* in the direction of motion. N2L in direction of motion gives – *F* = *ma* so $a = -\frac{F}{m}$, which is constant.	M1(AO3.3) A1(AO1.1)	Decide to use N2L to find acceleration No need to say 'constant'
		a	*B* (As *a* constant) use *suvat* , giving $S = 0 \times T - \frac{1}{2} \times \left(-\frac{F}{m}\right) T^2$ so $S = \left(\frac{F}{2m}\right) T^2$, and $k = \frac{F}{2m}$	B1(AO2.1) E1(AO2.4) [4]	Use appropriate (sequence of) *Suvat*
		b	As sliding, friction is limiting and *F* = *μR* *R* = *mg* $k = \frac{F}{2m}$ so $k = \frac{\mu mg}{2m}$ Hence $\mu = \frac{2k}{g} = \frac{2 \times 1.4}{9.8} = \frac{2}{7}$	M1(AO3.3) A1(AO3.4) M1(AO1.1) A1(AO2.2a) [4]	 In *F* = *μR*, substitute for *F* & *R* in terms of *m* and *g* Or 0.286 (3s.f.)
			Total	8	

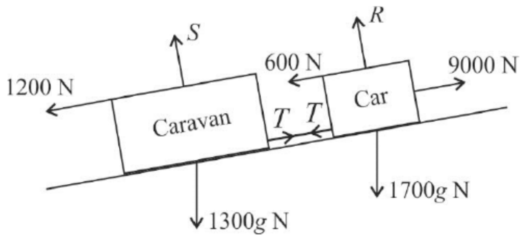
Question			Answer/Indicative content	Marks	Guidance	
15			Let T_1 and T_2 be the tensions in the strings to m kg mass and $2m$ kg mass respectively $T_1 = mg$ and $T_2 = 2mg$	B1(AO 3.3)	For values of tensions clearly stated or shown on the diagram	
			$T_2 - T_1 - F = 0$	M1(AO 3.3)		
			$F = mg$ towards the m kg mass (to the left on the diagram)	A1(AO 2.2a) [3]		
			Total	3		

Question			Answer/Indicative content	Marks	Guidance
16			$90 + 90 - 170 = 20a$	M1(AO 3.3)	For use of Newton II
			$a = 0.5$	A1(AO 1.1b)	
			$3 = 0 + \frac{1}{2} \times 0.5t^2$	M1(AO 1.1a)	For use of *suvat* with $s = 3$ and $u = 0$
			$t = \sqrt{12}$	A1(AO 1.1b)	3.464…
			$v = 0 + 0.5\sqrt{12}$	M1(AO 3.4)	For use of *suvat* with their a and t to find v
			$= \sqrt{3}$	A1(AO 1.1b)	1.732…
			In second phase resultant force is zero so $a = 0$	B1(AO 3.3)	Allow for 'constant speed' seen
			Time at constant speed $= \frac{7}{\sqrt{3}}$	M1(AO 1.1a)	
			Total time $= 2\sqrt{3} + \frac{7\sqrt{3}}{3} = 3.464\ldots + 4.041\ldots = 7.51$ s (3sf)	A1(AO 1.1b) [9]	
			Total	9	

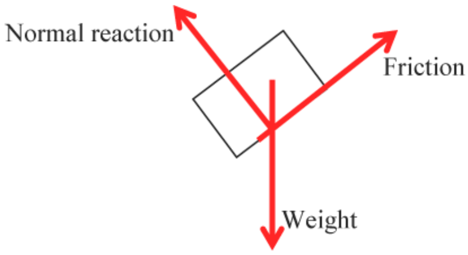
Question			Answer/Indicative content	Marks	Guidance	
17			Distance of 50 g piece from pivot is $(15 - x)$ cm	B1(AO 3.3)	soi; may be on the diagram	
			Moments about pivot: $20 \times 15 = 50 \times (15 - x)$ oe	M1(AO 3.1b)	Allow any consistent units of weight (or mass) and length mass) and length	Condone e.g. $20g$ with no units for g stated
				A1(AO 2.1)	Correct (unsimplified) equation	
			$x = 9$	A1(AO 1.1b)		
				[4]		
			Total	4		

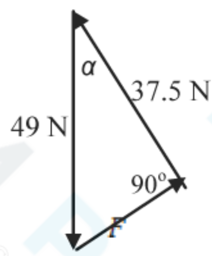
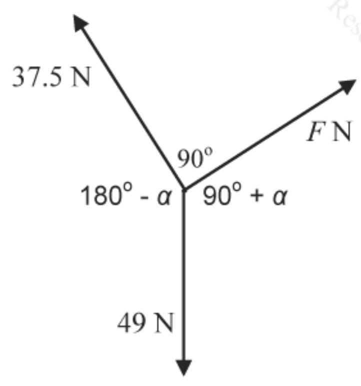
Question			Answer/Indicative content	Marks	Guidance
18			T_1 and T_2 are tensions in wire and string respectively; the globe has mass m		
			Resolve vertically: $T_1 \cos\theta = mg$	M1(AO 3.1b)	Allow sin/cos interchange for 1st M1 only
			$T_1 = \dfrac{mg}{\cos\theta}$		
			Resolve horizontally: $T_2 = T_1 \sin\theta$	M1(AO 3.3)	Use of trig identity
			$T_2 = \dfrac{mg}{\cos\theta} \times \sin\theta = mg\tan\theta$	M1(AO 2.1)	
			$mg\tan\theta < mg$ for $\tan\theta < 1$		
			$\theta < 45°$	A1(AO 2.2a)	Inequality clearly stated
			Alternative method		
			 Triangle of forces	M1	Forces shown, correct arrow directions
			Maximum T_2 is when $T_2 = W$	M1	Equating T_2 and weight for limiting case
			In this case $\theta = \arctan(1)$	M1	Use of trigonometry or isosceles triangle
			Hence $\theta < 45°$	A1	Inequality clearly stated
				[4]	
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
19		a	$21=\frac{1}{2}(u+0)14$ u = 3 AG	M1(AO 1.1a) A1(AO 2.1) [2]	Use of *suvat* equation(s) to find u
		b	$0 = 3^2 + 2a \times 21$ $a=-\frac{3}{14}$ Magnitude of force $=\lvert ma\rvert=18\times\frac{3}{14}$ $=\frac{27}{7}=3.86\text{ N}$	M1(AO 3.1b) M1(AO 1.1a) A1(AO 1.1b) [3]	Use of *suvat* to find acceleration Use of Newton II Must be magnitude (positive)
		c	$N = 18g$ $3.86 = \mu \times 18g$ μ = 0.022 to 2 sf	B1(AO 3.1b) M1(AO 3.4) A1(AO 1.1b) [3]	soi Use of $F = \mu N$ to find μ Must be given correct to 2sf
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
20		a			The two force diagrams may be shown separately
			Weights and normal reactions for each correctly shown	B1(AO 1.1a)	
			9000 N and force T in coupling correctly shown	B1(AO 3.3)	
			Resistances 600 N and 1200 N correctly shown	B1(AO 3.3)	
				[3]	
		b	Resolve weights parallel to the slope	M1(AO 3.1a)	Allow sin/cos errors here
			$9000 - 600 - 1200 = 3000g \sin 8 - 3000a$	M1(AO 3.3)	Newton II; all non-weight terms required
				A1(AO 1.1b)	All terms correct
			$7200 - 4091.689 = 3000a \Rightarrow a = 1.04$ m s^{-2}	A1(AO 1.1b)	cao
				[4]	

Question			Answer/Indicative content	Marks	Guidance
		c	$T - 1200 - 1300g \sin 8 = 1300 \times 1.04$ **or** $9000 - 1700g \sin 8 - 600 - T = 1700 \times 1.04$	M1(AO 3.4)	Resolving parallel to the slope for car or caravan; all forces present, correct mass
				A1(1.1a)	All terms in equation correct (for their *a*)
			$T = 4320$ N (answer using *a* unrounded)	A1FT(AO 1.1b)	FT their rounded / unrounded *a*
				[3]	
			Total	10	

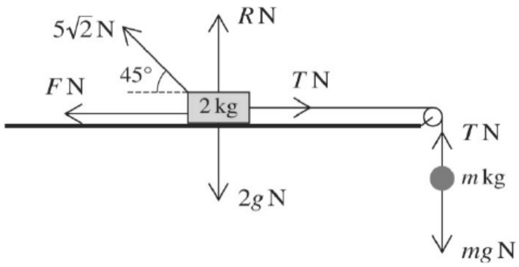
Question			Answer/Indicative content	Marks	Guidance
21		i	 	B1	**Forces** 3 correct forces.
				B1	**Labels** Accept 5*g* and *mg* for weight.
				B1	**Arrows**
					Missing force(s) If at least one of the 3 forces is missing, allow **SC1** for each fully correct force (ie including label and arrow) and ignore any additional forces that may be present.
					Extra force(s) Allow **B0** for **Forces** and up to **B1** for each of **Labels** and **Arrows** based on the correct forces and ignoring any extra(s).
					Components of weight Allow weight resolved into components parallel and perpendicular to the slope Accept **both** the weight and its components if the components are shown to be clearly different from the other forces (eg drawn with broken lines). Do not accept both the weight and its components if they all look the same; mark this as detailed under **Extra force(s)**.
				[3]	**Examiner's Comments** Part (i) of this question involved drawing a diagram for the forces acting on a block in equilibrium on a rough slope. While on the whole this was well answered, there were some surprising errors, such as showing the weight acting perpendicular to the slope or the normal reaction acting vertically. Some candidates showed the components of the weight as well as the weight itself; this is accepted if the components are presented differently from the other forces, for example using broken lines, but not if they all look the same.

Question			Answer/Indicative content	Marks	Guidance
		ii	$N = 5g\cos\alpha$ (or $37.5 = 5g\cos\alpha$)	M1	Do not allow sin-cos interchange
			$\cos\alpha = \frac{37.5}{49}$		
			$\alpha = 40.065...°$ so 40° to the nearest degree	A1	Must be rounded to 40°
			Frictional force = component of weight down the slope		
			$= 5g\sin 40.065...°$ (= 31.539…) so 31.5 N	B1	Allow any answer that rounds to 31.5 N Allow answer 31.5 N following two consistent sin-cos interchanges.
				[3]	
			Alternative Using a triangle of forces  	M1	Condone no arrows. Do not allow sin-cos interchange
			$\cos\alpha = \frac{37.5}{49} \Rightarrow \alpha = 40°$	A1	Must be rounded to 40°
			$= 5g\sin 40.065...°$ (= 31.539…) so 31.5 N	B1	Allow any answer that rounds to 31.5 N Allow answer 31.5 N following two consistent sin-cos interchanges.
			Alternative Using Lami's theorem  		
			$\frac{F}{\sin(180°-\alpha)} = \frac{37.5}{\sin(90°+\alpha)} = \frac{49}{\sin 90°}$	M1	

Question			Answer/Indicative content	Marks	Guidance
			$\alpha = 40°$	A1	Must be rounded to 40°
			$F = 31.5$N	B1	Allow any answer that rounds to 31.5 N Examiner's Comments In part (ii) candidates were required to use the given information to find the angle of the slope and the frictional force. Most did this successfully. However, the question asked for the angle to be given to the nearest degree and many lost a mark by giving it to some other level of accuracy.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
22		i	2×120×cos 20°–F = 430×0.05	M1	Newton's 2nd law, including *ma* term, friction and resolved force(s); allow sin-cos interchange for this mark only.
				A1	All terms and signs correct
			F = 204(.206…)	A1	
				[3]	**Examiner's Comments** In part (i) the forces acting on a sledge and its acceleration were given and the question asked for the force of resistance. This was answered correctly by nearly everyone.
		ii	430×a = –240.026…	M1	Apply FT from their F from part (i) throughout this part. All forces present
			a = 0.08366…	A1	Condone 0.08 for this mark
			Percentage increase is $\frac{0.0836...-0.05}{0.05}\times 100 \quad (=67.32...)$	M1	There must be evidence of a complete method for finding percentage change. The denominator must be the original acceleration and the original value must be subtracted from the new value at some stage.
			Percentage increase is 67.3% (to nearest 0.1)	A1	To allow for rounding and truncation, allow answers between 66% and 68% inclusive following otherwise correct working.
				[4]	**Examiner's Comments** In part (ii) candidates were asked to find the new acceleration and the percentage increase when the forces were applied in a different manner. Almost all found the new acceleration correctly but there were quite a lot of errors working out the percentage, for example using the wrong denominator and forgetting to subtract the original value. Teachers using this question in the classroom may like to compare the percentage increases in the forward force on the sledge (6.4%) and the acceleration (67.3%) and consider why there is such a large difference between them.

Question			Answer/Indicative content	Marks	Guidance
			Total	7	
23			Vertical equilibrium: weight = 10 + 15 = 25 N Moments about A: 25x = 15 x 0.8 x = 0.48 m	B1(AO1.1a) M1(AO3.1b) A1(AO1.1b) [3]	soi or take moments about other points
			Total	3	
24			Resolve down the slope, using Newtons' second law $mg \sin 20° = ma$ $v^2 = 0^2 + 2(g \sin 20°)s$ $v = \sqrt{13.407...} = 3.66 \text{ m s}^{-1}$	M1(AO3.3) A1(AO1.1b) M1(AO1.1a) A1(AO1.1b) [4]	Use of $v^2 = u^2 + 2as$
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
25		a		B1(AO1.1a) B1(AO3.3) [2]	Weights and normal reaction labelled Tension correct in both parts of the string and friction in the correct direction	Condone absence of units N in diagram Friction force could be shown as μ*R* or 0.6*R* and tension could be shown as *mg*
		b	$R+5\sqrt{2}\sin 45° = 2g$ $R = 2g - 5$ (= 14.6) $F = 0.6(2g - 5)$ $F = 8.76$ N	M1*(AO3.4) A1(AO1.1b) M1(AO1.1a)dep* A1(AO1.1b) [4]	Resolve vertically for the block Numerical evaluation not needed here Use of $F = \mu R$, but do not allow if $R = 2g$	Allow sin or cos here
		c	$T = F + 5\sqrt{2}\cos 45°$ $(=13.76)$ $T = mg$ $m = \dfrac{13.76}{g} = 1.40$	M1(AO3.4) A1(AO1.1b) M1(AO1.1a) A1(AO1.1b) [4]	Resolve horizontally for the block Correct equation; evaluation not required for this mark Resolve vertically for the ball	Allow sin or cos here soi
			Total	10		

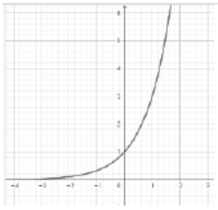
Question			Answer/Indicative content	Marks	Guidance
26		a	10 (m s^{-2})	B1(AO 3.4) [1]	
		b	*g* varies according to location	B1(AO 1.2) [1]	

Question			Answer/Indicative content	Marks	Guidance	
		c	$v = (u_1i + u_2 j) - 10tj$	M1(AO3.1b)	Differentiation of r to find v	
			Maximum height when $u_2 - 10t = 0$	M1(AO3.1b)	Equating their j component to zero	
			$t = \frac{u_2}{10}$	A1(AO1.1b)		
			$(u_1\mathbf{i} + u_2\mathbf{j})\frac{u_2}{10} - 5\mathbf{j}\left(\frac{u_2}{10}\right)^2 = 14\mathbf{i} + 20\mathbf{j}$	M1(AO1.1a)	Equate r with their *t* to given vector	
			$\frac{u_2^2}{10} - \frac{5u_2^2}{100} = 20 \Rightarrow u_2 = 20$	A1(AO1.1b)	cao, from equating j components	
			$u_1 \times \frac{20}{10} = 14 \Rightarrow u_1 = 7$	M1(AO3.1b)	Equating i components to find u_x	
			Initial velocity is (7i + 20j) m s^{-1}	A1(AO2.5)	Must be in vector form	Accept $\begin{pmatrix}7\\20\end{pmatrix}$
			Alternative solution			
			Vertical motion has $u = u_2$, $v = 0$, $a = -10$, s = 20	M1		
			$0 = u_2^2 + 2 \times (-10) \times 20$	M1	Use of *suvat* equation(s) leading to u_2	
			$u_2 = 20$	A1	cao	
			0 = 20 – 10t	M1	Use of *suvat* equation(s) leading to *t*	
			t = 2	A1		
			Horizontal motion: $14 = u_1 \times 2$ &⇒ $u_1 = 7$	M1	Use of their *t* in constant speed eqn	
			Initial velocity is (7i + 20j) m s^{-1}	A1 [7]	Must be in vector form	Accept $\begin{pmatrix}7\\20\end{pmatrix}$

Question			Answer/Indicative content	Marks	Guidance
		d	$21 = 7t$ &⇒ $t = 3$ Height at $t = 3$ is $20 \times 3 - 5 \times 3^2$ $= 15$ m	M1(AO3.1b) M1(AO1.1a) A1(AO3.2a) [3]	Finding t from horizontal motion Use of $s = ut + \frac{1}{2}at^2$ with their t
			Total	12	

Question			Answer/Indicative content	Marks	Guidance
27		a	Component of weight down the plane $4.7g \sin 60^{\circ}$ Equilibrium equation $T = 4.7g \sin 60^{\circ}$ $= 39.889\ldots$ so $T = 39.9$ to 3 sf	 B1 (AO 2.1) E1 (AO 3.3) [2]	AG Award if seen Must be clear that 39.9 N is the tension and not just component of weight **Examiner's Comments** This was generally well answered but some candidates lost a mark as all they had written was a component of weight and it was not clear that they had equated that to the tension. Since the answer was given in the question, the response needed to be a full mathematical justification to show the given answer. **AfL** Make sure that you set up an equilibrium equation even if there are only two terms.

Question			Answer/Indicative content	Marks	Guidance	
		b	Resolve perpendicular to the slope N is the normal reaction between plane and block B $N = 4g \cos 25^\circ$	B1 (AO 1.1a)	Need not be evaluated here [≈35.5]	
			Resolve up the slope	M1 (AO 3.3)	Allow only sign errors	
			$T - F - 4g \sin 25^\circ = 0$	A1 (AO 1.1b)	F need not be evaluated here [≈23.3]	
			On the point of sliding so $F = \mu N = \mu \times 4g \cos 25^\circ$	M1 (AO 3.1b)	Do not allow for $F \le \mu N$ unless = used subsequently. FT their values.	If only values are seen used, it must be clear that the values used are friction and normal reaction.
			$\mu = \dfrac{4.7g \sin 60^\circ - 4g \sin 25^\circ}{4g \cos 25^\circ} = 0.656$ to 3sf	A1 (AO 1.1b) [5]	FT (notice this answer is 0.657 if 39.9 used for T)	
					Examiner's Comments The problem solving element of this question stems from the lack of help that candidates were given in structuring their answer. They had to realise that they had to calculate the normal reaction and the frictional force before they could calculate the coefficient of friction. Some answers were very fragmented with very little help given by candidates to the examiner who were not always able to tell whether finding the normal reaction and the frictional forces had been attempted.	
			Total	7		

Question			Answer/Indicative content	Marks	Guidance
28		a	(i) 	B1 (AO 1.2) [1]	correct shape in both quadrants condone touching the *x*-axis, but not cutting it **Examiner's Comments** Sketches should make clear that the function tends towards $y = 0$ but does not cut the *x*-axis
			(ii) (0, 1)	B1 (AO 1.1) [1]	do not allow just $y = 1$ **Examiner's Comments** A clear indication of the coordinates (0,1) was required and not just $y = 1$.
		b	$(\mathrm{f}(x) =)\log_3 x$	B1 (AO 1.1) [1]	allow eg $\frac{\log x}{\log 3}$ **Examiner's Comments** There was an expectation that candidates would use base 3 logarithms, but a variety of correct functions were given and gained credit.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance	
29		a	$R = 25$	B1 (AO 1.1)		
			$\tan^{-1}\left(\frac{24}{7}\right)$ or $\sin^{-1}\left(\frac{24}{25}\right)$ or $\cos^{-1}\left(\frac{7}{25}\right)$	M1 (AO 1.1)		73.739795° rounded to 2 or more sf may imply **M1A0**
			$25\cos(x + 1.29)$	A1 (AO 1.1)	α= 1.28700221759 rounded to 2 or more sf	allow **A1** for α found to 2 or more sf
				[3]	<u>Examiner's Comments</u> The majority of candidates gained full credit, with careless arithmetic resulting in dropped accuracy marks on this routine item.	
		b	12 ± *their* 25	M1 (AO 3.1a)	or one of – 13 and 37 identified	
			$-13 \le f(x) \le 37$	A1 (AO 1.1)	allow eg from – 13 to 37 inclusive	**A0** if inequality is strict
				[2]	<u>Examiner's Comments</u> Some candidates answered their own question, taken directly from part (a).	
			Total	5		

Question			Answer/Indicative content	Marks	Guidance
30			$\begin{pmatrix} 8 \\ 0 \end{pmatrix} + \begin{pmatrix} 2a \\ -3a \end{pmatrix} + \begin{pmatrix} 0 \\ b \end{pmatrix} = 0$ $a = -4$, b = –12	M1 (AO2.5) A1 (AO1.1b) A1 (AO1.1b) [3]	Setting up a correct vector equilibrium equation or two separate equations. **Examiner's Comments** Many candidates did not form an equilibrium equation in vector form nor a pair of equilibrium equations for the two directions. Many made the mistake of writing $F_1 + F_2 = F_3$ or similar and received no marks. Others had correct equations to obtain the method mark but subsequent sign errors cost the accuracy marks. In future teaching, emphasise to candidates the importance of setting up their equilibrium equation i.e. total force = 0
			Total	3	
31			Equilibrium when P applied: $P - F = 0$ $\Rightarrow F = P$ Newton II when $3P$ applied: $3P - F = 5 \times 2$ So $2P = 10$, giving $P = 5$	B1 (AO 3.1b) M1 (AO 3.1b) A1 (AO 1.1) [3]	Friction force may appear as μR oe F in terms of P not needed here
			Total	3	

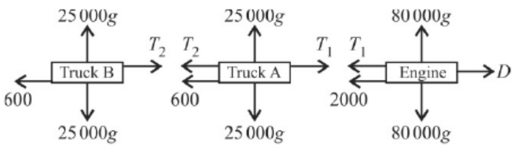
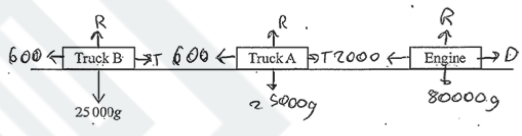
Question			Answer/Indicative content	Marks	Guidance	
32			Resolve →: $X - 45 = 0 \Rightarrow X = 45$	B1 (AO 1.1a)		
			Resolve ↑: $Y - 27 - 40 = 0 \Rightarrow Y = 67$	B1 (AO 1.1a)		
			Moments about A: $Xd - 27 \times 60 = 0$	M1 (AO 3.4)	oe; any moments equation must be correct, apart from sign errors	
			$d = \frac{27 \times 60}{45} = 36$	A1 (AO 1.1) [4]	FT their value for X	Answer for d must correspond to cm units; do not accept 0.36 or 0.36m.
			Total	4		

Question			Answer/Indicative content	Marks	Guidance
33		a	Resolve perpendicular to plane: $R = 2.7g \cos 25\,°$	B1 (AO 3.3)	Allow sin/cos interchange if consistent error elsewhere
			so $F = \mu R$ gives $F = 0.4 \times 2.7g \cos 25°$ (= 9.59…)	M1 (AO 3.4)	For 0.4 × their R, but not if $R = 2.7g$
				A1 (AO 1.1)	For correct expression for F; evaluation not needed here
			Newton II ∥ to plane: $24 - F - 2.7\,g \sin 25° = 2.7a$	M1 (AO 3.3)	All forces needed and the 'weight' term must be a component; allow sign errors
			$a = 1.1945…$ so acceleration is 1.19 m s^{-2} (3sf)	A1 (AO 1.1)	awrt 1.19
				[5]	
		b	After 5 s, $v = 0 + 1.1945 \times 5$	M1 (AO 3.1b)	Use of *suvat* leading to a value for v
			$v = 5.9725$	A1 (AO 1.1)	
			Newton II ∥ to plane: $-F - 2.7\,g \sin 25° = 2.7a$	M1 (AO 3.3)	Both forces, with weight resolved
			$a = -7.6943$	A1 (AO 1.1)	
			$v = 0$ at distance s where $0 = 5.9725^2 - 2 \times 7.6943 \times s$	M1 (AO 1.1a)	Use of *suvat* leading to a value for s
			$s = 2.3179…$ so distance travelled is 2.32 m (3sf)	A1 (AO 1.1)	cao
				[6]	
			Total	11	

Question			Answer/Indicative content	Marks	Guidance	
34			Vertical equilibrium $N = 5g$	B1 (AO 1.1a)	Must be seen – may be on a diagram	
			$F = P$	M1 (AO 1.1a)	May be implied	
			On the point of sliding $P_{max} = F[= \mu N] = 0.3 \times 5g$	M1 (AO 3.3)	Allow for $0.3 \times 5g$ even if B1 not awarded	
			$P_{max} = 14.7$	A1 (AO 3.4)	cao	
				[4]		
			Total	4		
35		a	Taking moments about A: $0.3 \times 100 + 0.75 \times 60$	M1 (AO 1.1b)	Allow for one moment only, but if two terms present, both must be moments	Do not allow for the total of a force and a moment
			= 75 Nm anticlockwise	A1 (AO 1.1b)	cao; direction needed	
				[2]		
		b	Minimum force when applied at B: $0.9F = 75$	M1 (AO 3.1b)	Forming equilibrium equation with distance 0.9	
			$F = 83.3$	A1 (AO 1.1b)	FT their moment in (a)	
				[2]		
		c	The hinge provides an additional [horizontal] force at A	B1 (AO 3.2a)	Must mention either A or the hinge	Do not allow either mark for an argument based on the weight
			But this will not have a moment about A	B1 (AO 3.2a)	Must mention moment about A	
				[2]		
			Total	6		

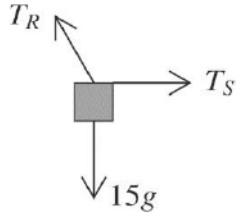
Question			Answer/Indicative content	Marks	Guidance	
36		a	Use of Newton's 2nd Law for the whole train Weight component parallel to the plane is $0.5g\sin 20°$ $2.5 - 0.5g\sin 20° = 0.5a$ $a = 1.65$ ms^{-2} to 3sf **Alternative solution** Equations for engine and truck separately N2L for engine: $2.5 - 0.3g\sin 20° - T = 0.3a$ N2L for truck: $T - 0.2g\sin 20° = 0.2a$ $a = 1.65$ms^{-2} to 3sf	B1 (AO 3.3) M1 (AO 1.1a) A1 (AO 2.1) B1 M1 A1 [3]	May be implied by use in an equation N2L with at least one correct force AG; must be clearly shown Correct weight component in at least one equation Both equations attempted, plus attempt to eliminate T AG; must be clearly shown	$m = 0.5$ used $m = 0.3$ and/or 0.2 used
		b	**Either** $T - 0.2g\sin° = 0.2 \times 1.65$ **Or** $2.5 - 0.3g\sin° - T = 0.3 \times 1.65$ $T = 1$ N	M1 (AO 3.3) A1 (AO 1.1b) [2]	Use of N2L for either engine or truck Allow awrt 1.00	Equation must have all the forces and no extras
		c	$3^2 = 1^2 + 2 \times 1.65 \times s$ $s = 2.42$ m	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Use of suvat leading to a value for s Or 2.43 from use of $a = 1.6482\ldots$	
			Total	7		

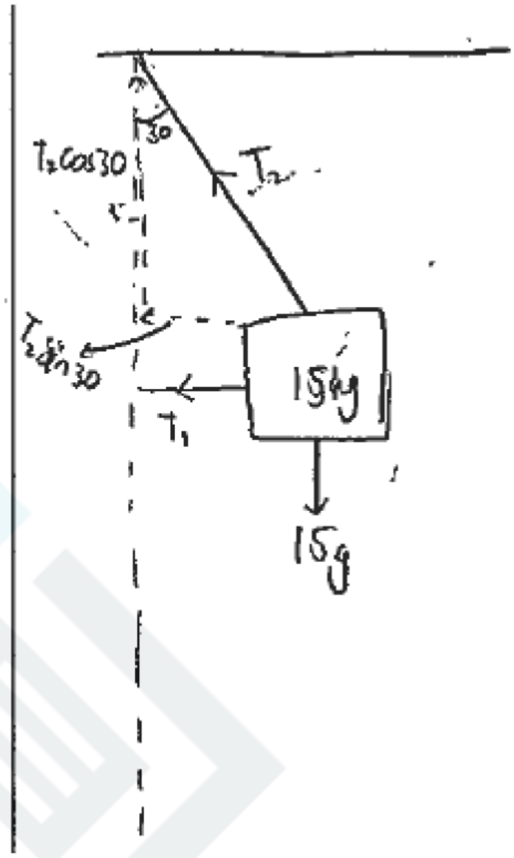
Question			Answer/Indicative content	Marks	Guidance
37	a		(W =) –2.5 g**j** N (= –24.5 **j** N)	B1 (AO1.2) [1]	Condone missing units. Must be negative. Allow any correct vector notation **Examiner's Comments** This had to be precise to obtain the mark and many candidates omitted at least one of the minus signs, the g or the **j**. As well as testing the accurate use of notation, the question here was structured to remind candidates to include the weight in their subsequent calculation.
	b		Newton's second law – 24.5**j** + (3**i** – 2**j**) + (– **i** + 18**j**) = 2.5**a** $\mathbf{a} = \frac{1}{2.5}(2\mathbf{i} - 8.5\mathbf{j}) = (0.8\mathbf{i} - 3.4\mathbf{j})\text{ms}^{-2}$	M1 (AO1.1a) A1 (AO1.1) [2]	Attempt to use N2L with m = 2.5 Condone missing weight or an incorrect weight vector — Must be sum of forces = 2.5a Cao – do not award if the magnitude of acceleration given as acceleration **Examiner's Comments** The method mark here was awarded for an attempt to use Newton's second law either in vector form or two separate equations in the two directions. Many candidates had an equation with a mixture of vector and scalar terms which was not given any credit
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
38				B1 (AO1.1a) B1 (AO1.1a) B1 (AO1.1a) [3]	Weights and normal reactions shown Tensions (must be two distinct) shown Driving force (D or 16 200) and three resistances correctly shown Allow N_1, N_1 and N_2 instead of values to match weights for the normal reaction (condone N_1, N_2, N_3) **<u>Examiner's Comments</u>** It was quite common to see both tensions labelled with T – the subsequent question even states that these are different. It was also common to omit the upwards vertical forces or to include them and label them all as R. Exemplar 1 Force diagram This diagram only has the tensions in the direction of motion and it was quite common to see both tensions labelled with T –the subsequent question even states that these are different. It was also common to label the upwards vertical forces all as R or omit them altogether.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
39	a		Component in **i** direction zero ⇒ $k = -4$	B1 (AO 3.3) [1]	Allow (-4**i** + 5**j**) seen instead of k | Do not allow for $k = -4$**i** or similar. <u>Examiner's Comments</u> Many candidates produced quite a bit of working for a 'Write down…' question, but many had the correct value. The mark was not given where k was given as a vector.

Question			Answer/Indicative content	Marks	Guidance	
	b		Weight = $-0.8g\mathbf{j}$	B1 (AO 1.2)	Allow if seen	
			N2L $5\mathbf{j} + 3\mathbf{j} - 0.8g\mathbf{j} = 0.8\mathbf{a}$	M1 (AO 1.1a)	Condone missing weight	
			[$\Rightarrow \mathbf{a} = 0.2\mathbf{j}$]			
			$\mathbf{v} = (4\mathbf{i} + 7\mathbf{j}) + 0.2\mathbf{j} \times 10$	M1 (AO1.1a)	Using their **a** in a vector *suvat* equation(s)	
			velocity is $(4\mathbf{i} + 9\mathbf{j})$ m s^{-1}	A1 (AO 2.5)	Must be in correct vector form	Accept fully correct column vector
			Alternative solution	[4]		
			Acceleration is vertical so consider only vertical motion	M1	Applying N2L in 1-dimension	
			$5 + 3 - 0.8g = 0.8a$	B1	Condone missing weight Including the weight (consistent sign convention)	
			[$\Rightarrow a = 0.2$]			
			$v = u + at = 7 + 0.2 \times 10 = 9$	M1	Using *suvat* equation(s) in the vertical direction	
			So the velocity is $(4\mathbf{i} + 9\mathbf{j})$ m s^{-1}	A1	Must be in vector form	Accept fully correct column vector
				[4]	<u>Examiner's Comments</u> There was considerable confusion about whether to use vectors or scalars and some candidates had equations with a mixture. More common was the omission of the weight in the N2L equation, resulting in an incorrect acceleration and eventually a very large vertical component of the velocity. Again, there was a lot of premature rounding of intermediate calculations, but there was some very efficient work from many candidates. Some realised that the fact there was a zero-horizontal component of force meant that they could work in a vertical direction until the end of the question, but the final answer had to be a vector.	
			Total	5		

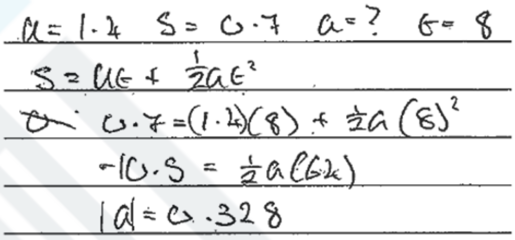
Question			Answer/Indicative content	Marks	Guidance
40	a			B1 (AO 1.1a) [1]	Three forces in approximately correct directions, with arrows and labels; tensions must be distinct Accept *W* or *mg* for the weight; condone missing *g* for this mark <u>Examiner's Comments</u> This question was mostly well done, but some diagrams omitted arrows. Some had two arrows on the strings, presumably because they believed that it was necessary to show the tension from both ends; if the arrow closest to the box was correct this was marked correct. A few had both forces on the same side of the box and occasionally other forces appeared (such as a normal reaction). The mark was also lost where *T* was used for both tensions, rather than inventing different names.

Question			Answer/Indicative content	Marks	Guidance
					Exemplar 3  This exemplar shows good practice in that the forces and the components of the forces used in later parts are shown very differently. Unfortunately, the horizontal force is in the wrong direction and the mark could not be awarded (although the candidate was able to achieve full marks in part (c) with a negative value for the tension).

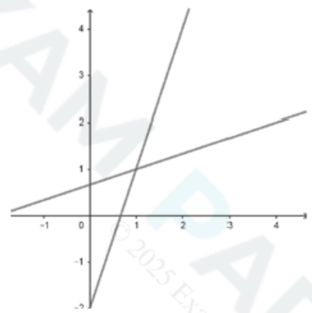
Question			Answer/Indicative content	Marks	Guidance
	b		Resolve vertically: $T_R \cos 30° = 15g$ $T_R = \dfrac{15 \times 9.8}{\cos 30°} = 98\sqrt{3} = 170 \text{ N (3sf)}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Forming equilibrium equation (allow sin/cos interchange) Oe **Examiner's Comments** This was mostly done correctly, but some candidates interchanged sine and cosine either in error or by marking the wrong angle 30°. Some candidates resolved the weight rather than the tension, creating an incorrect equilibrium equation.
	c		Resolve horizontally: = $T_S = T_R \sin 30°$ T_S = 84.9 N (3sf)	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Allow sin / cos interchange if consistent with (b) Oe Accept $T_S = 15g \tan 30°$ **Examiner's Comments** This was also done well by those candidates who had a good diagram in part (a).
			Total	5	

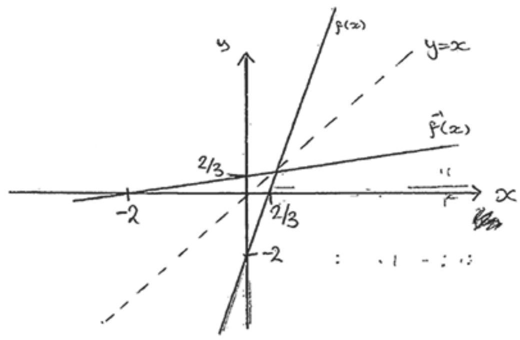
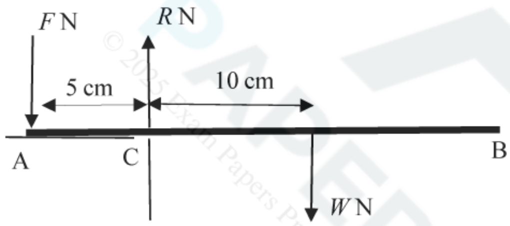
Question			Answer/Indicative content	Marks	Guidance	
41	a		Resolve down the plane $2g \sin 20° = 2a$ $a = g \sin 20°[= 3.352]$	M1 (AO 3.1b) A1 (AO 3.3)	N2L with no other forces with attempt to resolving the weight Allow $a = -g \sin 20°$ if it is clear that up the slope is the positive direction.	Allow sin / cos interchange for M mark
			$s = ut + \frac{1}{2}at^2$ with $s = 0.7$, $u = 1.4$ and their a $0.7 = 1.4t + \frac{1}{2}(g \sin 20°)t^2$	M1 (AO 3.3)	Using *suvat* equation(s) leading to an equation for t; allow sign errors Correct equation using their $0.5a$	
			$(4.9\sin 20°)t^2 + 1.4t - 0.7 = 0$	A1 (AO 3.4)	Allow coefficient of t^2 1.7 or better	Solution BC is sufficient
			Time taken is 0.352 s	A1 (AO 3.2a)	Cao Must select correct root if two roots given	Allow if positive root only seen
				[5]	<u>Examiner's Comments</u> Most candidates realised the need to resolve in the direction of motion and there were plenty of fully correct answers. A few candidates tried to resolve horizontally and vertically, but usually had incomplete or incorrect equation so their working did not lead to a solution. AfI In a question like this it is expected that candidates use their calculator to solve the quadratic.	

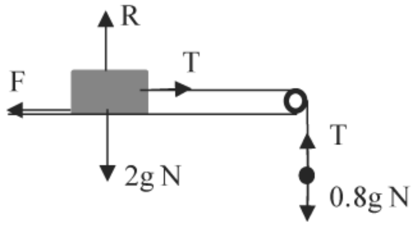
Question			Answer/Indicative content	Marks	Guidance
	b		Friction (and / or air resistance) would have the effect of slowing the particle so ignoring friction underestimates time	B1 (AO 3.5a) [1]	Needs to indicate why ignoring resistance produces an underestimate — eg “friction will slow it down” for B1 **<u>Examiner’s Comments</u>** Candidates need to give a proper explanation here; just stating that there is a friction force is not enough, the mark was for explaining that the resistance would result in a slowing of the particle. **AfI** In a question starting ‘Explain…’ a response stating friction is present is not enough. The link between force information and time taken also needs to be explained, with a reference to acceleration or velocity in some way.

Question			Answer/Indicative content	Marks	Guidance	
	c		*s* = 0.7, *u* = 1.4, *t* = 0.8 ⇒ 0.7 = 1.4 × 0.8 + $\frac{1}{2}a \times 0.8^2$	M1 (AO 3.3)	Use of *suvat* equation(s) to find *a*; allow sign errors	
			$a = \frac{0.7 - 1.12}{0.32}$ = –1.3125 (–1.31 to 3 sf)	A1 (AO 1.1b)	Allow *a* = 1.3125 if sign convention clear	eg both *u* and *s* negative
				[2]	<u>Examiner's Comments</u> This was well answered, but some candidates were uncomfortable with the negative value they obtained and tried to change that. **Exemplar 6**  u = 1.4 s = 0.7 a = ? t = 8 s = ut + ½at² ⇒ 0.7 = (1.4)(8) + ½a(8)² −10.5 = ½a(64) |a| = 0.328 This exemplar shows the application of the misread rule. It is clearly stated that the value to be used is 8 rather than the 0.8 in the question. It also shows how the sign of the acceleration has been disregarded (some candidates just ignored the minus); this candidate did at least use modulus notation, before assuming that the acceleration must be positive because the particle is moving down a slope. Candidates who changed *a* =–1.3125 to 1.3125 were not given the A mark.	

Question			Answer/Indicative content	Marks	Guidance	
	d		N2L down the plane $2g \sin 20° - F = 2 \times (-1.3125)$ ($F = 9.32859…$)	M1 (AO 3.1b) A1 (AO 1.1b)	All forces present and weight resolved; allow sin / cos interchange Fully correct equation; F need not be evaluated here	Notice R may be found first. Allow sin / cos interchange in *their* first equation but the second equation must be consistent to award M1
			Resolve perpendicular to the plane $R = 2g \cos 20°$ ($R = 18.4…$) Use of $F = \mu R$ $\mu = \frac{9.32859...}{18.4...} = 0.506$ (3sf)	M1 (AO 3.3) A1 (AO 1.1b) M1 (AO 3.4) A1 (AO 1.1b) [6]	No extra forces Fully correct equation; R need not be evaluated here Allow for their F and R used Accept 0.506 or 0.507 Must be 3sf <u>Examiner's Comments</u> Many fully correct solutions were seen, with candidates able to resolve in two directions and manipulate the algebra to obtain a value for the coefficient of friction. Misconception Candidates need to be aware that the F in the equation $F = ma$ and the F in the equation $F \leq \mu R$ are not the same. It can be better to teach Newton's second law as 'resultant force = *ma*' rather than use the shorthand equation.	
			Total	14		

Question			Answer/Indicative content	Marks	Guidance
42	a		$y = 3x - 2$ so $x = \dfrac{y+2}{3}$ $\mathrm{f}^{-1}(x) = \dfrac{x+2}{3}$	M1 (AO1.1a) A1 (AO1.1) [2]	Condone '*y* =' but eg '*f*(*x*) = ' does not score **<u>Examiner's Comments</u>** This question provided an accessible start to the paper for most candidates with most scoring all 6 marks. A few candidates did not understand the concept of an inverse function and instead gave the reciprocal of f(*x*) in this part.
	b			B1 (AO1.1) B1 (AO1.1) [2]	$y = 3x - 2$ going through (0, –2) Or $y = \dfrac{x+2}{3}$ going through (0, 2/3) $y = \mathrm{f}^{-1}(x)$ a reflection of $y = \mathrm{f}(x)$ in $y = x$ — Line of symmetry roughly at 45° or implied by coords Both graphs linear to get B2 **<u>Examiner's Comments</u>** Most candidates provided good quality sketches and understood the need for the inverse to be a reflection in $y = x$; the best solutions added $y = x$ to their sketches. While a sketch does not need scaled axes, 'important points' should be marked such as the intercepts.

Question			Answer/Indicative content	Marks	Guidance	
					Exemplar 1  This candidate clearly shows the line of symmetry making it easier for them to draw the graphs and the examiner see their intent. They clearly indicate the y-intercepts and x-intercepts as per the notes in the specification about the level of detail required for a sketch.	
			Total	4		
43	a		 	B1(AO1.1a) B1(AO1.1) [2]	F and their weight even if not labelled in roughly correct position Distances labelled. Allow for 5cm **and** any measurements that show that the weight is acting in the centre of the ruler.	R is already drawn in Printed Answer Booklet
	b		Take moments about C $10 \times 0.028g = 5F$ $F = 0.549$ N	M1(AO1.1a) A1(AO1.1) [2]	Moments about any point with all relevant forces seen. Allow their weight and one incorrect distance allow $\frac{7}{125}g$ oe	[Value of R is 0.8232 N]
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
44	a			B1(1.1a) B1(1.1a) B1(1.1a) [3]	both weights correct. common tension in the rigtht directions Friction and normal reaction and no extra forces	Allow weight of box and weight of sphere, but not if both marked weight. Allow T_1 and T_2 provided they are clearly shown equal to each other elsewhere. For *F,* allow 0.35R, 0.35 × 2 *g* 0.7g or 6.86 N
	b		$T - F = 2a$ $0.8g - T = 0.8a$	B1(AO1.1a) B1(1.1a) [2]	Allow any expression for F Allow distinct tensions if consistent with diagram	For *F,* allow 0.35*R*, 0.35 × 2*g* 0.7g or 6.86 N
	c		Vertically for the block $R = 2g$ Friction $F = 0.35R = 0.7g$ Add equations $0.8g - F = 2.8a$ $0.8g - 0.7g = 2.8a$ $a = 0.35$ m s^{-2}	M1(AO3.1b) A1(AO2.1) M1(AO2.1) A1(2.1) [4]	Attempt to use μ to evaluate friction Correct value for F Eliminate *T* from their equations AG must follow from correct work	Some of this work may already have been seen in previous part
	d		Use $s = 0.5$, $u = 0$, $a = 0.35$ $0.5 = \frac{1}{2} \times 0.35 \times t^2$ $t = 1.69$ s	M1(AO1.1a) A1(AO1.1) [2]	Using *suvat* equation(s) leading to a value for *t* Do not allow ±1.69	
			Total	11		

Question			Answer/Indicative content	Marks	Guidance	
45	a		**W** = (–*mg* sin 30°)**i** + (–*mg* cos 30°)**j** $\left[\mathbf{W} = \left(-\frac{1}{2}mg\right)\mathbf{i} + \left(-\frac{\sqrt{3}}{2}mg\right)\mathbf{j}\right]$	M1(AO3.1b) A1(AO2.5) [2]	Attempting to resolve the weight. Allow sin / cos interchange and sign errors for the method mark. All correct in this vector form	*mg* must be seen for the method mark.
	b		**W** + **R** + **F** = 0	B1(AO2.5) [1]	Allow any rearrangement of this. Allow if their expression for **W** is used instead of **W**	
	c		**R** = *R***j**	B1(AO3.1b)	Allow for any clear indication that **R** is a multiple of **j** or that it has no component in the **i** direction	May be implied with an equation for the i direction with two terms and an equation in the j direction with three terms
			[(-*mg* sin 30°) **i** + (-*mg* cos30°) **j** + (6**i** + 8**j**) + *R***j** = **0**]	M1(AO3.1a)		
			i component –*mg* sin 30° +6 = 0	A1(AO1.1)	Forming equation from their **i** terms, or equivalent by resolving parallel to the plane. FT their **W**	
			giving *m* = 1.22 to 3 sf	A1(AO1.1)	Correct equation in **i** direction **AG**	
			j component: –12cos 30 + 8 + *R* = 0	M1(AO3.1a)	Equation from the **j** terms (must include all three terms), oe, and using value of *m*	12 is the value for *mg*. 1.22 × 9.8 = 11.956
			R = 2.39 so magnitude is 2.39 N	A1(AO3.2a) [6]	Accept arwt 2. 4	
			Total	9		

Question			Answer/Indicative content	Marks	Guidance	
46	a		weight $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix}$	B1(AO1.2)	Allow seen or implied by correct answer	
			Equilibrium equation $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \mathbf{F}_2 = \mathbf{0}$	M1(AO1.1a)	FT their weight. Allow sign errors	
			$\mathbf{F}_2 = \begin{pmatrix} -4 \\ 2+1.5g \end{pmatrix} = \begin{pmatrix} -4 \\ 16.7 \end{pmatrix}$ N	A1(AO1.1) [3]	must be vector allow **i-j** form	
	b		Newton's second law $\begin{pmatrix} 0 \\ -1.5g \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 20 \end{pmatrix} = m\mathbf{a}$	M1(AO1.1a)	Addition of vectors. Allow if weight missing but other two forces and acceleration seen in equation	Allow wrong weight only if given as a vector
			$\mathbf{a} = \frac{1}{1.5}\begin{pmatrix} 6 \\ 3.3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2.2 \end{pmatrix}$ m s^{-2}	A1(AO1.1) [2]	Must be vector; any correct form.	
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
47	a		[h(x) or fg(x) =] $\sqrt{x^3 - x - 6}$ oe $x > 2$	B1 (AO1.1) B1 (AO1.1) [2]	expression domain	mark the final answer
	b		√18 oe isw FT *their* h(x)	B1 (AO1.1) [1]	allow 4.2426406872… rounded to 2 or more sf	
	c		$\frac{1}{2}\times\frac{3x^2-1}{\sqrt{(x^3-x-6)}}$ or $\frac{3x^2-1}{2\mathrm{h}(x)}$ *their* $\frac{\mathrm{dh}}{dx}$ evaluated at $x = 3$ $\frac{3\sqrt{2}}{13}$ or 0.326356975932 rounded to 2 sf or better *OR* $x^2 = y^3 - y - 6 \Longrightarrow 2x\frac{dx}{dy} = 3y^2 - 1$ oe $\frac{dx}{dy} = \frac{3y^2-1}{2x}$ *or* $\frac{dy}{dx} = \frac{2x}{3y^2-1}$ substitution of $y = 3$ and $x =$ *their* $\sqrt{18}$ $\frac{3\sqrt{2}}{13}$ or 0.326356975932 rounded to 2 sf or better	M1 (AO3.1a) A1 (AO1.1) M1 (AO1.1) A1 (AO3.2a) [4] M1 A1 M1 A1 [4]	chain rule used all correct **Examiner's Comments** Generally this question was well received and proved to be a good discriminator. Candidates who did well scored full marks in parts (a) and (b) and knew how to approach part (c) – although they may have made a slip in differentiation or in the substitution. Candidates who did less well were not able to apply the chain rule successfully in part (c), or substituted incorrect values in the penultimate step. Candidates who did not do well incorrectly simplified their expression in part (a) and	allow one slip in differentiation, eg sign error h(x) must be correct for first **M1** allow one slip eg sign error rearrangement to find h–1(x) explicitly in terms of x followed by differentiation does not score

Question			Answer/Indicative content	Marks	Guidance
					were unable to make meaningful progress in part (c), although they may have earned credit for a correct domain in part (a) and a FT calculation in part (b). **Misconception** A significant minority of candidates appeared unclear how composite functions are formed and thought fg(x) = f(x) × g (x). There was also a surprising number of candidates that expressed $\sqrt{x^3 - x - 6}$ as $\sqrt{x^3} - \sqrt{x} - \sqrt{6}$
			Total	7	